Control VAR: a counterfactual based approach to inference in macroeconomics

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Abstract

This paper addresses the challenges of giving a causal interpretation to vector autoregressions (VARs). I show that under independence assumptions VARs can identify average treatment effects, average causal responses, or a mix of the two, depending on the distribution of the policy. But what about situations in which the economist cannot rely on independence assumptions? I propose an alternative method, defined as control-VAR, which uses control variables to estimate causal effects. Control-VAR can estimate average treatment effects on the treated for dummy policies or average causal responses over time for continuous policies.

The advantages of control-based approaches are demonstrated by examining the impact of natural disasters on the US economy, using Germany as a control. Contrary to previous literature, the results indicate that natural disasters have a negative economic impact without any cyclical positive effect. These findings suggest that control-VARs provide a viable alternative to strict independence assumptions, offering more credible causal estimates and significant implications for policy design in response to natural disasters.

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1 Introduction

What would have happened to inflation had central banks around the world not introduced unconventional monetary policy on the offset of the Great Financial Crisis? Is monetary policy effective in controlling inflation? What is the impact of monetary policy on the economy?

While some of these questions may appear to be too tough to tackle, economists have them frequently in mind, and find themselves at loss as to what methods to use to answer them. Long run, short run, sign restrictions in VARs, LPs, or linear regressions are just some of the hundreds tools specifically developed to address these issues. Yet, economists find themselves in front of the difficult reality of being unable to provide a proper causal answer to their questions. A lot of such issues would naturally involve the formulation of a counterfactual. When we think about the effectiveness of unconventional monetary policy, we have in mind a counterfactual world in which the central banks could have preferred inaction over unconventional monetary policy in the face of a looming disinflationary pressure. When we think about the effects of natural disasters, we have in mind a counterfactual world in which no disaster happened. Yet, we rarely make the case that we are truly identifying the effect of unconventional monetary policy over inaction; or the effect of the disaster. For economists, talking about causality in macroeconomics has for a long time been a forbidden topic.

This is an issue that exists almost exclusively in social sciences. The lack of the possibilities of running experiments leaves economists with the goal of making informed guesses. **?** make the comparison of economists and weather forecasters, painting a gloomy picture of economic science. They, however, express optimism towards the direction social sciences have been aiming for.

This could be because, recently, much of the literature has focused on the development of tools that would allow economists to make truly causal claims that involve the formulation of a counterfactual in models akin to the one of **?** for tools such as VARs or Local Projections, with a particularly relevant progress coming from the works of **?**, **?**, and **?**. Another stream of literature has instead taken the road of developing methods for the definition of clear counterfactuals by making use of definitions of optimal policies (**?**). Both these literatures have, at heart, the goal of clearly defining the effects of macroeconomic policies as the difference between two alternative assignments: one in which a causing variable is zero, one in which it is one.

This paper has the goal of providing economists with additional tools to make truly causal claims. Identification in macroeconomics frequently uses Impulse Response Functions (IRF) to analyze short and long-run effects of changes in a policy variable on an outcome variable. This is a particularly needed feature when it comes to data with sufficiently large time components, since having an idea of possibly long-term effects of a change in a policy can result in valuable information. Suppose the economist is trying to identify the effect of natural disasters on the US economy. One commonly utilised approach would be the one proposed by **?**. They estimate a VAR that includes the cost of natural disasters (CD), Industrial Production (IP), and Macroeconomic Uncertainty (MU). They argue that a recursive identification scheme could be appropriate to identify the impact of natural disasters on the US economy.

From the point of view of the causal literature, this example could be the perfect one to test the theoretical framework of **?**. They show that if the natural disasters are (1) independent on the past of other policies, (2) the future of other policies, (3) other contemporaneous policies; (4) and are independent to the potential outcomes of industrial production and macroeconomic uncertainty;[1] VARs are causal. They could potentially identify the Average Treatment Effect (ATE): the average effect of a natural disaster hitting the US economy.

This paper therefore tries to understand what type of causal effects can be identified under the model of **?** under that specific example.

Several important issues arise. First, the majority of natural disasters in the United States happened in August and September, which could violate the assumption of independence (4) if there are hidden vulnerabilities or strengths specific to that period of time of the US economy.[2] Second, because Industrial Production and Macroeconomic

[1]Essentially, natural disasters do not happen in systematic coincidence with economic fluctuations.

[2]Everything that is not captured by seasonality could potentially introduce a selection bias because

Uncertainty are non-stationary, a filter is required. This creates an issue because filters will contaminate the informativeness of the estimator, an issue also recognised by **?**, which call those effects F-ATE. Third, the distribution of the treatment variable - the cost of natural disaster - unevenly weights some disasters over others (see **?**). This last feature has been found to be particularly problematic by the broader causal literature. I show that, in this case, the effects will be a mix of an ATE and an ACRT, similarly to **?**. This is particularly relevant since non-negative policies are often the object of causal inference and the complexity of the interpretation of a mix of two causal effects can be troublesome.

Perhaps the simplest case of a VAR that could claim causality ends up having features such as seasonality, filters, or discontinuities in the distribution which heavily impact its validity and interpretation. Essantially, independence assumptions are stronger than many of the identification schemes currently utlised by the literature, even if the link from estimator to estimand may not be as clear as the one they describe. [3]

As an alternative to such methods, I present a novel approach to estimating causal effects that uses some control series that share a trend with the outcomes to eliminate the long-run component from the inference problem through cointegration, a system that will take the name of CVAR (Control VAR). Such an idea can be implemented in two different ways. The simplest one is to estimate a VAR on the difference of the treated and control variable. The second approach is to simply estimate a Vector Error Correction Model (VECM) that includes the control variables. Such estimation methods will therefore be named *simple difference* and *VECM* respectively.

These approaches have several advantages compared to the wide range of independence assumptions previously described. First, they do not make use of any independence or exogeneity assumption. Second, they do not require any form of filtration[4]. Third, they only require the user to discuss whether the assumptions are satisfied, and have a

it is unobserved.

[3]For instance, economists frequently use instruments - not recursive identification - to identify the effect of monetary policy because even if there is not a widely understood concept of SVAR-IV capturing a LATE, they know independence would not hold.

[4]This advantage is particularly relevant if one considers that most macroeconomic time series are non-stationary (see **?**).

known estimation that is readily available in any statistical toolkit.

In the simple difference approach, the assumptions needed are fairly intuitive. Suppose we use German macroeconomic uncertainty as a counterfactual for the United States. Because the two series are cointegrated, they share common long-run trends and thus provide valuable low-frequency information to the researcher. The approach requires two conditions: first, that in the absence of a natural disaster, U.S. uncertainty would have followed the same low-frequency path as observed in Germany; and second, that in normal times, the two trends move together without diverging structurally. These assumptions are demanding, especially the first. For example, while Hurricane Katrina had devastating effects on U.S. uncertainty, Germany was not completely insulated from spillover effects on its own level of uncertainty.

For this reason, the paper takes a second approach to cases in which the simple difference assumptions may not be satisfied.

The VECM approach uses lags in place of the original variables. Essentially, say now that the researcher runs a VECM with uncertainty in the United States and in Germany. In the equation for macroeconomic uncertainty in the US at time t, german data is indexed at time $t-1$. In this case, the assumption required is that the common trend is sufficient to eliminate long run structural variations in the United States.

Under these assumptions, if the policy variable is binary, the CVAR identifies an Average Treatment effect on the Treated (ATT). If the policy is normally distributed it is possible to identify an ACR.

The IRFs estimated by CVAR may appear to be substantially different from the IRFs estimated without controls. In the case of no controls, the IRF may present long-term effects. In the case of the CVAR, the IRF may converge to zero within short periods because the long run component is eliminated by the control series (**?**). This is exactly what this paper finds in its empirical application.

I show that the original system of **?** estimates the effect of a shock in disasters to be negative on IP for two months and indefinitely positive after that. The natural disaster shock impact is positive on MU for three months and negative for the remaining two years.[5]

As an alternative, I estimate a CVAR through a VECM using the same variables and adding German Industrial Production and Macroeconomic Uncertainty as a control. I use German data because out of the countries for which a MU series is available (see **?**), Germany is the one that provides a higher rank statistic. To identify an average treatment effect on the treated, I shift from a continuous and non-negative variable (cost of natural disasters) to a dummy equal to one when the cost of such disasters is above the 95-*th* quantile. Therefore, the estimated effect is an ATT representing the difference between large and non-existent or minor disasters. [6]

The resulting Impulse Response Functions present a much shorter horizon of relevance when compared to the results of the original paper. This suggests that the control based approach may solve the classic issue of possibly long-term effects of temporary shocks when the outcome variables are non-stationary (**?**).

The discussion will be organized as follows.

Section 2 introduces the notation and definitions required to discuss the nature of the causal effects in the context of VAR. Section 3 focuses on the first contribution of the paper, the causal interpretation of a VAR. Subsection 3.1 discuss the case of a dummy policy variable, a result drawn from **?** but framed under my alternative causal notation. Subsection 3.2 discusses the case of a normally distributed policy variable. Subsection 3.3 discusses the case of a non-negative continuous policy variable. Section 4 focuses on the second contribution of the paper. It introduces the CVAR, and is divided between subsection 4.1, which introduces the simple difference approach, and subsection 4.2, which discusses the VECM approach. Section 5 discusses the empirical applications and gives a context for the results. Section 6 concludes.

[5]This is an extension of the result of the original model which plots logistic transformations of 6 horizon shocks.

[6]Appendix A.0.3 presents similar results obtained using all the available countries from the **?** dataset.

2 Definition of causal effects

2.1 Notations and definitions

The economist is generally interested in estimating a dynamic causal effect of assignments on outcomes in observational time series settings. Every time period, a unit receives a vector of assignments, and their outcomes can be observed. There exists a causal relation between the assignments and the outcomes through a *potential outcome process*, a stochastic process that generally defines the counterfactual assignment paths. A *dynamic causal effect* is typically described as the difference of potential outcome processes along distinct assignment paths at a specific moment in time.

The problem for the economist is related to the inherent impossibility of observing both the potential outcomes under treatment and the potential outcomes in the absence of a treatment at a given time. This problem is usually dealt with by inferring a *counterfactual* which represents what would have been the value of the outcome variable, if the treatment variable was zero when the treatment was one. Many ways of inferring such values have been introduced across different scientific fields and types of available data and experiments.

Time series methods have traditionally relied on the computation of a counterfactual using the forecast of the outcome variable. This is the case, among others, of intervention analysis (**?**) and Structural VARs (**?**).

I will consider the following model of the outcome variable for the identification of causal effects:

$$Y_{j,t}(w_{j,k}) = Y_{j,t}(W_{j,1:t-1}, W_{j,1:k-1,t}, w_{j,k}, W_{j,k+1:K,t}, W_{j,t+1:t+h}),$$

is the potential outcome process of the outcome variable $Y_{t,j}$ under the assignment $w_{j,k}$, where $k = 1,..,K$ is an index that indicates multiple possible treatments and $j = 1,..,J$ indicates the generic outcome variables. For example, if the researcher is interested on the causal effect of hurricanes and heat waves on GDP, inflation, and unemployment, $K = 2$

and $J = 3$. Therefore $W_{j,k,t}$ describes the amount of treatment $k \in \{1, ..., K\}$ received by series $j \in \{1, ..., J\}$ at time $t \in \{1, ..., T\}$, with $w_{j,k,t}$ denoting one realization from the random variable. Moreover, this definition highlights that the potential outcome depends on the past of the policies, their future, and the contemporaneous policies that are not the target of analysis.

With such notation, a dynamic causal effect can be defined as $Y_{j,t}(w_{j,k,1:t}) - Y_{j,t}(w'_{j,k,1:t})$. Often Impulse Response Functions that derive from a VAR are interpreted as $Y_{j,t}(w_{j,k,t}) - Y_{j,t}(w'_{j,k,t})$, which means that the interest is not on the effect of receiving a subsequent set of treatments $w_{j,k,1:t}$ versus no treatments $w'_{j,k,1:t}$, but an artificial type of counterfactual that captures the effect of a non time dependent policy variable. For the sake of simplicity, $w_{j,k}$ will be considered a 1% shock and $w'_{j,k}$ will be considered a 0% shock. This substantially simplifies the notation.

Only recently **?** highlighted some restrictive conditions under which the impulse responses may be given a causal interpretation. They discuss two cases: (i) $Y_{j,t}$ is stationary, (ii) $Y_{j,t}$ is not stationary. According to their interpretation, in the first case the economist may identify an ATE, defined as $\mathbb{E}[Y_{j,t}(1) - Y_{j,t}(0)]$, if the sources of selection bias are independent on the treatment. This means that the treatments are required to be independent of their history and their future, on other contemporaneous treatments, and on the potential outcome path of the outcome variable. In the case of non-stationary variables, this needs to hold conditionally on the filtration, so that the causal effect is defined as $\mathbb{E}[Y_{j,t}(1) - Y_{j,t}(0)|\mathcal{F}_{t-1}]$.

However, more causal effects could be imputed to Vector Autoregressions.

For example, a VAR identified using external instruments can be interpreted as a LATE (**???**). **?** analyzes causality in panel vector autoregressions and shows that their interpretation depends on the distribution of the policy variable and the assumptions the economist may put on the data and the associated narrative. In that case, however, the system assumes the existence of a large unit component, which may not be available in the case of macroeconomic data.

Hence, this paper takes two aims:

1. clarify the causal interpretation of VAR with macroeconomic data, and highlight the different causal effects that can be identified using VAR with different policy distributions under the same type of independence assumptions as **?**,
2. propose using controls to eliminate the long run component of time series data through cointegration, a method named CVAR.

2.2 Causal effects

The potential outcome for series j at time t can be written as

$$Y_{j,t}\Big(W_{k,1:J,1:T}\Big),$$

indicating a dependence on the treatments received by all series at all times, both past and future. We restrict this dependence under the following assumptions:

Assumption 1 (SUTVA)**.** *The potential outcomes for series j depend only on its own treatment path. That is,*

$$Y_{j,t}\Big(W_{k,1:J,1:T}\Big) = Y_{j,t}\Big(W_{k,j,1:T}\Big).$$

See also **?**.

Assumption 2 (Homogeneity)**.** *The treatment paths are identical across units, so that*

$$Y_{j,t}\Big(W_{k,j,1:T}\Big) = Y_{j,t}\Big(W_{k,1:T}\Big).$$

Assumption 3 (Absence of Anticipation)**.** *Future treatments do not influence present outcomes. Formally, the potential outcome at time t depends only on the treatment path up to t:*

$$Y_{j,t}\Big(W_{k,1:T}\Big) = Y_{j,t}\Big(W_{k,1:t}\Big).$$

Under the Rubin's causal framework, I define the following assumption, which binds the observed and potential outcomes.

Assumption 4. *(Realized outcomes): The realized outcomes are linked to the potential outcomes through the following assignment process:*

$$Y_{j,t}^{obs} = \begin{cases} Y_{j,t}(1) & \textit{if } W_{j,k,t} = 1 \\ Y_{j,t}(0) & \textit{if } W_{j,k,t} = 0 \end{cases}$$

This definition binds the observed outcome at time t to depend on the value of the policy variable. It means that if the policy variable corresponds to a treatment status 1, then we observe the potential outcome of the variable Y_t under the treatment status. If the policy variable corresponds to a non-treatment, then the observed outcome variable can be only observed under a non-treatment. A counterfactual is therefore defined as what we do not observe, that is:

Assumption 5. *(Non realized outcomes): The non realized outcomes are linked to the potential outcomes through the following assignment process*

$$Y_{j,t}^{nobs} = \begin{cases} Y_{j,t}(0) & \textit{if } W_{j,k,t} = 1 \\ Y_{j,t}(1) & \textit{if } W_{j,k,t} = 0 \end{cases}$$

Remark 1. Counterfactuals can only be generated my making assumptions about the non-observed treated unit. For example, the DiD literature regularly assumes that the potential outcome of the treated units under no treatment $\mathbb{E}[Y_t(0)|W_{j,k,t} = 1]$ can be credibly formulated by using information from the control units.

According to assumptions 4 and 5 it is possible to define the causal effects. The definitions that follow indicate the outcome variable in the first part of the parentheses, the potential outcome in the second one, and the conditioning argument in the third one.

An ATE represents the difference of the outcome variable under a treatment 1 and the absence of a treatment 0.

$$\text{ATE}_{j,k}(Y_j|1,0) = \mathbb{E}[Y_j(1) - Y_j(0)].$$

An ATE therefore represents a measure that is close to the ideal of a dynamic causal effect, since it captures the potential outcome value under the treatment 1 versus the treatment 0. For example, it could represent the effect of a natural disaster versus no natural disaster.

An ATT is defined as the same difference, with the conditioning argument that the times observed are only ones in which the treated unit is subject to the treatment, i.e.

$$\text{ATT}_{j,k}(Y_j|1,0|1,1) = \mathbb{E}[Y_j(1)|W_{j,k,t} = 1] - \mathbb{E}[Y_j(0)|W_{j,k,t} = 1].$$

An ATT is generally the result of inference conducted using a direct counterfactual, so that the treated units are compared to control units. It may represent the effect of a natural disaster versus no natural disaster, under a conditioning argument that depends on the time extracted to be one in which there was a natural disaster.

I will also consider the ACR, Average Causal Response, which represent the effect of marginal changes in the treatments on the outcome variable. The concept of ACR was first discussed by **?**, which make the case for the advantages of a more comprehensive measure of causal effects that may not be dummies. Hence, it considers the effect of variations in the value of $W_{j,k,t}$ on the outcome variable $Y_{j,t}$,

$$\text{ACR}_{j,k}(Y_j|w_{j,k}) = \mathbb{E}\left[\frac{\delta Y_j(w_{j,k})}{\delta w_{j,k}}\right],$$

where $w_{j,k}$ is a value of a continuous $W_{j,k,t}$. Ideally, the researcher would be able to identify the impact of moving along different values of the treatment variable $W_{j,k,t}$ without assuming linearity. In practice, however, linearity is frequently a useful assumption in cases in which the data is not sufficiently dense around some target values. If linearity holds, the ACR is the most informative quantity the economist could capture. One way to think about the ACR is akin to the most traditional definition of the linear regression coefficient under the assumptions of exogeneity, and its derivative form makes it possible to generate credible counterfactual comparisons. For example, through the ACR, the economist could answer questions regarding the difference between the effect of a natural disaster that causes damages for 1000 billion dollars versus 100 billion dollars. This is possible through the comparison of the predicted values of the outcome variable under a plug-in quantity of the policy variable. They also have a conditional counterpart, defined as ACRT.

$$\text{ACRT}_{j,k}(Y_j|w_{j,k}|w_{j,k}) = \mathbb{E}\left[\frac{\delta Y_j(w_{j,k})}{\delta w_{j,k}}|W_{j,k,t} = w_{j,k}\right].$$

In this case, such quantity includes an additional conditioning argument that is related to the time being extracted being equal to $w_{j,k}$. In this case, the ACRT implements a selection bias that depends on the treatment time. For example, if natural disasters are more frequent in some periods of the year, the ACRT indicates that the esitimates of the impact of natural disasters on the economy will be biased by the fact that natural disasters are concentrated in specific seasons. There are conditions under which it is possible to move from the ACRT to the ACR, and therefore pin the selection bias to zero. The most common one is the independence of the treatment assignments to the potential outcomes.

Finally, all such causal effects may have conditional counterparts which may be useful in cases in which the outcome variable is non stationary. Hence, the filtered causal effects are as follows.

1. $\text{F-ATE}_{j,k}(Y_j|1,0) = \mathbb{E}[Y_j(1) - Y_j(0)|\mathcal{F}_{t-1}]$,
2. $\text{F-ATT}_{j,k}(Y_j|1,0|1,1) = \mathbb{E}[Y_j(1)|W_{j,k,t} = 1, \mathcal{F}_{t-1}] - \mathbb{E}[Y_j(0)|W_{j,k,t} = 1, \mathcal{F}_{t-1}]$,
3. $\text{F-ACR}_{j,k}(Y_j|w_{j,k}) = \mathbb{E}\left[\frac{\delta Y_j(w_{j,k})}{\delta w_{j,k}}|\mathcal{F}_{t-1}\right]$,
4. $\text{F-ACRT}_{j,k}(Y_j|w_{j,k}|w_{j,k}) = \mathbb{E}\left[\frac{\delta Y_j(w_{j,k})}{\delta w_{j,k}}|W_{j,k,t} = w_{j,k}, \mathcal{F}_{t-1}\right]$.

Remark 2. The literature is ambiguous about the interpretability of any filtered treatment effect. In particular, the macroeconometrics traditional approach has been to first difference or filter the series before estimating a VAR to avoid violations of the Wold theorem. The appeal to such procedures is evident. Making any type of causal claim while using a non-stationary variable would be impossible due to the non-finiteness of their moments and filtering and first differencing avoids the burden of stationarity. However, it is important to recognize the limitations of any kind of causal estimation procedure that results from a process of filtration, since it adds an additional form of uncertainty.[7]

3 Identification using VAR

A VAR can be identified in several ways. In the case of non-stationarity, the most commonly utilized procedure is including all the variables in a first differenced vector of the type that follows.[8]

$$\Delta X_t = (\Delta W'_{1,t}, .., \Delta W'_{j,k,t}, \Delta Y'_{1,t}, .., \Delta Y'_{J,t})'. \tag{1}$$

The goal of the economist is to estimate a causal effect for each combination of the policies with every outcome variable. However, the general idea of VAR models is that there are autoregressive components which may need to be modelled. Hence, the VAR equation is as follows.

$$\Delta X_t = \sum_{l=1}^{p} A_l \Delta X_{t-l} + \epsilon_t \tag{2}$$

Here, $\epsilon_t = (\widetilde{W}'_1, .., \widetilde{W}'_k, \widetilde{Y}'_1, .., \widetilde{Y}'_J)'$ is the vector of errors. Normally the causal effects are represented as the contemporaneous effect of each variable within X_t. This is represented through the matrix A_0.

$$A_0 \Delta X_t = \sum_{l=1}^{p} A^*_l \Delta X_{t-l} + \epsilon^*_t, \quad \epsilon_t \sim \mathcal{D}_p(0, \Omega). \tag{3}$$

[7]The first uncertainty comes from the filtration procedure. The second from the AR estimation. The third from the way the variance-covariance matrix of the residuals is utilized to carry inference. Often standard confidence intervals only embed the second type of uncertainty.

[8]The results of this section hold even in the case of a VECM estimated in the original variables.

Here $A_l^* = A_0 A_l$ for all l, $\epsilon_t^* = A_0 \epsilon_t$ and $\epsilon_t^* \sim \mathcal{D}_p^*(0, \Omega^*), \Omega^* = A_0^* \Omega A_0^{*\prime}$, where $\mathcal{D}$ is a generic distribution (not necessarily normal).

Example 1. Consider the case of one outcome variable and one policy variable, so that $j = 1$ and $k = 1$. In that case,

$$\Omega = \begin{bmatrix} var(\widetilde{W}_t) & cov(\widetilde{W}_t, \widetilde{Y}_t) \\ cov(\widetilde{W}_t, \widetilde{Y}_t) & var(\widetilde{Y}_t) \end{bmatrix}$$

is the variance-covariance matrix used for the causal estimation. The associated Choleksy decomposition is

$$chol(\Omega) = \begin{bmatrix} \sqrt{var(\widetilde{W}_t)} & 0 \\ \frac{cov(\widetilde{W}_t, \widetilde{Y}_t)}{\sqrt{var(\widetilde{W}_t)}} & \sqrt{var(\widetilde{W}_t) - (\frac{cov(\widetilde{W}_t, \widetilde{Y}_t)}{\sqrt{var(\widetilde{W}_t)}})^2} \end{bmatrix} = \begin{bmatrix} o_{11} & 0 \\ o_{12} & o_{22} \end{bmatrix}.$$

Inference is usually carried out by normalizing $chol(\Sigma_E)$ through an array $(1/\sqrt{var(\widetilde{W}_t)}, 0)'$, which makes the response to a unitary shock in $\widetilde{W}_t$ become $(1, \gamma)'$, where $\gamma = \frac{cov(\widetilde{W}_t, \widetilde{Y}_t)}{var(\widetilde{W}_t)}$.

One important message can be drawn from the example. Inference is, in general, conducted on $\widetilde{Y}_t$, and not directly on Y_t. This is particularly relevant because an argument for making use of identification through VAR is that, if the disturbances are symmetrically distributed, the estimation error of the autoregressive coefficients goes to zero. Such argument has been alternatively put forth by **?** in Theorem 2.1 or **?** in footnote 15. It has also created some debates about whether GLS may be a better alternative than the traditional 2 step procedures for the estimation of VAR since there is no guarantee that the errors are going to be symmetric (see **?**).

Remark 3. An alternative to the Cholesky decomposition could be put forth in the case of a not autocorrelated policy variable. In this case, an alternative system of the type

$$\Delta Y_t = \sum_{l=1}^{p} A_l \Delta Y_{t-l} + \epsilon_t,$$

where $\Delta Y_t = (\Delta Y_{1,t}', .., \Delta Y_{J,t}')$. Then, inference can be carried out by simply running a set of linear regressions of the disturbances on the policy variables $W_{j,k,t}$[9]. While the following discussion will focus on the first case, this second type of system could provide some advantages as it may be more parsimonious in the number of estimated coefficients. Moreover, while all the results that follow will be discussed in terms of the distribution of the innovation of the policy variable, they can be easily extended to be representative of the effects estimated using $W_{j,k,t}$, the distribution of the original policy variable.

Remark 4. While other procedures for the identification of causal effects, such as short run, long run, and sign restrictions, have been proposed in the macroeconometrics literature, I do not discuss the interpretation of such methods since they often result in a system of partially identified equations. This complicates the interpretation of the estimands and analyzing the resulting set of impulse responses would require an ad-hoc type of causal system that admits uncertainty over the assignment process.

The discussion that follows will focus on three different distributions of the policy variable, and highlight the assumptions required to conduct inference. Those cases have been selected because they may represent likely scenarios in which to carry causal inference. Nevertheless, from a purely mathematical perspective, any kind of feasible combination of policy distribution/assumption can result in a different causal estimand. For example, a policy that is distributed with a known distribution that is not normal can still result in a causal estimand that includes a weight times a derivative.

3.1 Dummy variable

Consider the case of a simple linear regression in which the independent (policy) variable is a dummy that indicates the treatment status of the dependent (outcome) variable. This is the case discussed by **??**. It can easily be shown that γ_{jk} estimates effects that are equivalent to the difference in mean estimator between $\widetilde{Y}_{j,t}(\widetilde{W}_{j,k,t} = 1)$ and $\widetilde{Y}_{j,t}(\widetilde{W}_{j,k,t} = 0)$, if the following assumption is satisfied.

[9]Basically this system would consist on (1) estimating the VAR, (2) running a set of $K \times J$ regressions, (3) plugging the results in the covariance matrix and using them to generate the Impulse Response Functions.

Assumption 6. *(Policy is a dummy variable) At each time $t \geq 1$, either $\widetilde{W}_{j,k,t} = 1$ or $\widetilde{W}_{j,k,t} = 0$.*

Assumption 7. *(Independence of treatments) For each k and j, the reduced form shock $\widetilde{W}_{j,k,t}$ is independent from*

1. the other policies, i.e. $\widetilde{W}_{j,k,t} \perp \widetilde{W}_{j,1:k-1,t}$ and $\widetilde{W}_{j,k,t} \perp \widetilde{W}_{j,k+1:K,t}$
2. its past, $\widetilde{W}_{j,k,t} \perp \widetilde{W}_{j,k,1}, .., \widetilde{W}_{j,k,t-1}$
3. its future, $\widetilde{W}_{j,k,t} \perp \widetilde{W}_{j,k,t+1}, .., \widetilde{W}_{j,k,t}$
4. the potential outcome process, $\widetilde{W}_{j,k,t} \perp \widetilde{Y}_{j,t}(\widetilde{W}_{j,k,1:t})$

Assumption 7 states that the assignment to the treatment must be independent of other policies ($\widetilde{W}_{j,-k,t}$), the past of all the policies, the future of all the other policies, and whether it happened because it relates to a particular potential outcome of the outcome variable. This last assumption is the reason why such an approach is almost never used in macro economics. For example, monetary policy is especially targeted to have an effect on the state of the economy by means of the Taylor rule (**?**). The resulting ATE theorem is:

Theorem 1. *(Identification of ATE) Under assumptions 6 and 7, the Cholesky decomposition of a VAR identifies*

$$\gamma_{j,k} = \mathbb{E}[\widetilde{Y}_j(1) - \widetilde{Y}_j(0)] = ATE_{j,k}(\widetilde{Y}_j|1,0)$$

It should be noted that the ATE in this theorem is the target of a lot of the research conducted in economics. In particular, the claim in the original work of **?** is that the ATE is always the target of Structural Vector Autoregression. However, most macroeconomic shocks, such as fiscal or monetary policy, are not measured by dummy variables. For example, the first is measured in terms of government spending or taxes, and the second in terms of changes in interest rates. Neither is measured in terms of a dummy variable that credibly represents a 1% shock versus a 0% shock. Accordingly, subsection 3.2 will discuss a more credible scenario in which the policy variable is continuously distributed. In the case of non-stationary filtered outcome variable the estimated effect becomes an F-ATE$_{j,k}(\widetilde{Y}_j|1,0)$.

3.2 Normally distributed continuous variable

Often the researcher is interested in the effects of a policy that is not a dummy variable that indicates treatment periods, and the most common alternative is the one of the normality of the errors distribution. I will show that the resulting estimand, in the case of the same assumption as 7, is an ACR.

First, consider the following assumptions, which bind the distributions of the policy innovations to be normal and the outcomes innovations to be continuously differentiable.

Assumption 8. *(Normality) The reduced form disturbances are normally distributed around zero, so that, for every $t \geq 1$ and $k \geq 1$, $\widetilde{W}_{j,k,t} \sim \mathcal{N}(0, \sigma_1^2)$.*

Assumption 9. *(Continuous differentiability) The reduced form innovations of the outcome variable $Y_{j,t}$, for every $j \geq 1$ and every $t \geq 1$, are continuously differentiable with respect to $W_{j,k,t}$ for any $k \geq 1$.*

Theorem 2. *(Identification of ACRT) Under assumptions 8 and 9, the Choleksy decomposition of a VAR identifies*

$$\gamma_{j,k} = \int q(w_{j,k}) g_j'(w_{j,k}) dw_{j,k} = \frac{\delta \mathbb{E}[\widetilde{Y}_j(w_{j,k})|\widetilde{W}_{j,k,t} = w_{j,k}]}{\delta w_{j,k}} = ACRT_{j,k}(\widetilde{Y}_j|w_{j,k}|w_{j,k})$$

where $q(w_{j,k}) > 0$ and $\int q(w_{j,k}) dw_{j,k} = 1$. The weights are:

$$q(w_{j,k}) = \frac{1}{\sigma^2_{\widetilde{W}_k}} \int_{-\infty}^{\infty} (\mathbb{E}[\widetilde{W}_k] F_{\widetilde{W}_k}(w_{j,k}) - \theta_{\widetilde{W}_k}(w_{j,k}))$$

and $\Theta_{\widetilde{W}_k}(w_{j,k}) = \int_{-\infty}^{w_{j,k}} m f_{\widetilde{W}_k}(m) dm = F_{\widetilde{W}_k}(w_{j,k}) \mathbb{E}[\widetilde{Y}_j(w_{j,k})|\widetilde{W}_k = w_{j,k}]$.

Therefore, according to theorem 2, the ACRT gives information about the entire distribution of causal effects of the policy variable on the outcome variable. However, it still includes a conditioning component which could contaminate inference. Under the same type of independence assumptions as **?**, it can be shown that it becomes the ACR.

Theorem 3. *(Identification of ACR) Under assumptions 7, 8, 9, the Cholesky decomposition of a VAR identifies*

$$\frac{\delta\mathbb{E}[\widetilde{Y}_j(w_{j,k})]}{\delta w_{j,k}} = ACR_{j,k}(\widetilde{Y}_j|w_{j,k}).$$

A few considerations are in order. First, it is possible to eliminate the selection bias by using the same set of independence assumptions that are used by **?**. Second, this is not the only case in which it is possible to move from the ACRT to the ACR. Later in the paper (in section 4.1.2) I will show that it is possible to obtain a similar type of quantity under strict conditions on counterfactual units.

The following paragraph will show that deviations from the normality assumption, in particular in the case of a non-negative distribution, can be harmful.

In the case of non-stationarity, the estimated effects on the filtered outcome become respectively an F-ACRT$_{j,k}(\widetilde{Y}|w_{j,k}|w_{j,k})$ and an F-ACR$_{j,k}(\widetilde{Y}|w_{j,k})$ depending on whether the economist relies on an independence type of assumption.

Remark 5. In the case of a non-normal but symmetric distribution, it is possible to recover the ACRT and ACR by making use of carefully selected weights. This is partially discussed in **?** and belongs to a stream of the biometrics and microeconometrics literature that dates back to **??**.

3.3 Non-negative continuous policy variable

Often the researcher is interested in the effects of a non-negative continuous policy. This is, for example, the effect of natural disasters measured by their damage on the economy.

A recently developed stream of research has evolved around isolating the kind of effects that such policies may identify. This case is frequently defined as "continuous treatment". In a TWFE with a continuous non-negative policy the effects identified are a weighted sum of an ATE and an ACRT (see **?**). While it is possible to generate scenarios and restrictions to bind such causal effects to a more interpretable quantity, it is particularly hard to do so in macro economics due to the lack of credible restrictions. For example, the micro econometrics literature makes the argument that, by using controls, such effects could be disentangled, especially using non-parametric types of estimators. Since the focus of the attention of this paper is on VARs, this result will be considered especially negative because it implicitly shows that to obtain sensible causal estimates the researcher would need to make use of non-parametric estimators, which is a type of estimator that is rarely utilized in the context of Vector Autoregressions.

Assumption 10. *(Non-negative policy) For each t, the policy of interest k is such that $W_{j,k,t} \geq 0$.*

Theorem 4. *(Identification of ATE+ACR) Under assumptions 7, 9, 10, the Cholesky decomposition of a VAR identifies*

$$\gamma_{j,k} = \int_{d_L}^{d_U} q_1(w_{j,k})(ACRT_{j,k}(\widetilde{Y}_j|w_{j,k}) + \frac{\delta ATT(\widetilde{Y}_j|w_{j,k}, w_{j,k}|a, 0)}{\delta a}|_{a=w_{j,k}}) + q_0 \frac{ATE_{j,k}(\widetilde{Y}_j|d_L, 0)}{d_L}$$

where

$$q_1(w_{j,k}) := \frac{\mathbb{E}[\widetilde{W}_k|\widetilde{W}_k \geq w_{j,k}] - \mathbb{E}[\widetilde{W}_k])\mathbb{P}(\widetilde{W}_k \geq w_{j,k})}{var(\widetilde{W}_k)} \quad and \quad q_0 := \frac{(\mathbb{E}[\widetilde{W}_k|\widetilde{W}_k > 0] - \mathbb{E}[\widetilde{W}_k])\mathbb{P}(\widetilde{W}_k > 0)d_L}{var(\widetilde{W}_k)}$$

The conclusion that can be drawn from Theorem 4 is that the identified causal effects are a mix of different distributions with weights that depend on the density of the policy variable, and specifically the frequency of it being above zero and/or above $w_{j,k}$. The general conclusion is that if the weights are non-negatively distributed, the estimated effects will hardly be interpretable.

In the case of non-stationarity of the outcome variable, the estimated effects become

$$\int_{d_L}^{d_U} q_1(w_{j,k})(\text{F-ACRT}_{j,k}(\widetilde{Y}|w_{j,k}) + \frac{\delta F - \text{ATT}(\widetilde{Y}_j|w_{j,k}, w_{j,k}|a, 0)}{\delta a}|_{a=w_{j,k}}) + q_0 \frac{\text{F-ATE}_{j,k}(\widetilde{Y}|d_L, 0)}{d_L}.$$

4 The CVAR approach

The results of the previous section may appear to be troublesome. In particular, I have highlighted some relevant problems in the causal macroeconomics literature:

1. To obtain quantities like the ATE and the ACR, the required assumptions on VARs are unrealistic, since we need to advocate the independence of the treatment on their past, future, other policies, and the potential outcomes.

2. Even in the most advocated case of normally distributed residuals and independence assumptions, the causal effects could be difficult to interpret if normality is violated.

This section will consider a different type of approach to causal inference that is inspired from the DiD literature. The idea is to use one or many control variables to provide an alternative to causal inference procedures that suffer from the first two of the problems above, a setting that I will define CVAR (Control VAR).

I will discuss two ways by which controls can be implemented in structural vector autoregressions. The first method will essentially discuss the case of a VAR estimated on the difference between the treatment and control outcome variable, that will therefore be named *simple difference*. Its interpretation is closely connected to the one of DiD, but its estimation presents some disadvantages:

1. It is vulnerable to the high probability of spillover effects when dealing with macroeconomic variables.

2. It does not allow for the estimation of a rescaling coefficient that relates the control and treated variable, which means it assumes a 1:1 relationship between the trends of the treated and controls.

3. It can only be estimated with one control variable.[10]

[10]This problem could be solved by using a weighted average of the control variables, with a procedure similar to synthetic controls (**??**). This type of approach, while possibly convenient, could require a strict set of assumptions on the estimation of weights such as the presence of a common data generating process in the absence of the intervention in a factor model type of system.

In the second part, I will propose a method/assumption scheme that may appear to be more credible in macroeconomic setting, which will make use of standard VECM estimation procedures to estimate the CVAR. This alternative approach will therefore be defined as *VECM*.

4.1 Estimation through simple difference

Define $X_t = (W'_{1,t}, .., W_{j,k,t}, (Y^1_{1,t} - Y^0_{1,t})', .., (Y^1_{J,t} - Y^0_{J,t})')'$, where the superscript is 1 for treated units and 0 for the non treated ones. Under the following alternative system, estimate the VAR:

$$X_t = \sum_{l=1}^{p} X_{t-l} + \epsilon_t,$$

and run the Cholesky decomposition on ϵ_t. First, I will describe the assumption required for the identification procedure in the case of a dummy policy in section 4.1.1. Then, I will move to the case of a continuous policy in section 4.1.2.

4.1.1 Dummy policy in the simple difference case

I will use the following assumptions:

Assumption 11. *The observed outcome of the treated and untreated units is*

$$Y_t^{1,obs} = \begin{cases} Y_{j,t}(1) & \text{if } \widetilde{W}_{j,k,t} = 1 \\ Y_{j,t}(0) & \text{if } \widetilde{W}_{j,k,t} = 0 \end{cases} \qquad Y_t^{0,obs} = \begin{cases} Y_{j,t}(0) & \text{if } \widetilde{W}_{j,k,t} = 1 \\ Y_{j,t}(0) & \text{if } \widetilde{W}_{j,k,t} = 0 \end{cases}$$

and the non observed outcome of the treated units is

$$Y_t^{1,nobs} = \begin{cases} Y_{j,t}(0) & \text{if } \widetilde{W}_{j,k,t} = 1 \\ Y_{j,t}(1) & \text{if } \widetilde{W}_{j,k,t} = 0 \end{cases}$$

Assumption 12. *(Difference stationarity): For each $j \geq 1$, $k \geq 1$, $t \geq 1$, $Y^1_{j,t} - Y^0_{j,t}$ is stationary.*

Assumption 13. *(Deviations from the controls in treated times are causal). For each $j \geq 1$, $k \geq 1$,$t \geq 1$,*

$$\mathbb{E}[Y^1_{j,t}(0) - Y^0_{j,t}(0)|\widetilde{W}_{j,k,t} = 1] = 0.$$

Assumption 14. *(Deviations from controls in non-treated times are zero): For each $j \geq 1$, $k \geq 1$,$t \geq 1$,*

$$\mathbb{E}[Y^1_{j,t}(0) - Y^0_{j,t}(0)|\widetilde{W}_{j,k,t} = 0] = 0.$$

Assumption 11 binds the control unit to always be not treated, and allows the potential outcome of the treated unit to depend on their treatment status. Moreover, it assumes that the potential outcomes are common to the units, so that the potential outcomes of the non-treated units under a not treatment scenario, and the one of the treated units under a non-treatment scenario, are the same. Assumption 12 prevents violations of the Wold theorem. Assumption 13 is used similarly to a parallel trend condition. It means that the unobserved outcome of the treated units in treated time can be reasonably compared to the realized one of the control units. Assumption 14 is similar to a no anticipation condition, in that the units are also assumed to have similar behavior in the untreated times after conditioning on their past difference.

Theorem 5. *(Identification of ATT in the direct control case). For each $j \geq 1$, $k \geq 1$,$t \geq 1$, if the policy variable is a dummy and assumptions 11, 12, 13, and 14, are satisfied, the Cholesky decomposition of the direct control approach of the CVAR identifies*

$$\gamma_{j,k} = \mathbb{E}[Y_{j,t}(1) - Y_{j,t}(0)|\widetilde{W}_{j,k,t} = 1] + \Delta_{AR} = ATT_{j,k}(Y_j|1,0|1) + \Delta_{AR},$$

where

$$\Delta_{AR} = \mathbb{E}[Y^1_{j,t} - Y^0_{j,t}|\widetilde{W}_{j,k,t} = 0, Y^1_{j,t-1,j} - Y^0_{j,t-1}, ...] - \mathbb{E}[Y^1_{j,t} - Y^0_{j,t}|\widetilde{W}_{j,k,t} = 1, , Y^1_{j,t-1,j} - Y^0_{j,t-1}, ...]$$

is a residual component that depends on the prediction of the VAR autoregressive components.

The presence of a residual component in the interpretation of the causal estimand should not necessarily be an element of concern. It indicates the presence of a selection bias that depends on the prediction of the autoregressive component of the VAR model that becomes zero in the case of common predictions of the treated and control units. One way to interpret this additional component is to think about it as the idea that treated and control variables may have common autoregressive components in equilibrium, and deviations from such long term equilibrium come from the residual impact of interventions, which negatively weights the causal estimand. Symmetrically, it is negatively impacted by potential spillover effects.

Most importantly, the theorem implies that using a control variable can result in a causal estimand without making use of any exogeneity assumption.

4.1.2 Continuous policy in the simple difference case

In the case of a continuous policy, the CVAR estimated through the direct control approach will require assumption 12 and the following assumptions to hold:

Assumption 15. *The observed outcome of the treated and untreated units*

$$Y_t^{1,obs} = \begin{cases} Y_{j,t}(w_{j,k}) & \textit{if } \widetilde{W}_{j,k,t} = w_{j,k} \\ Y_{j,t}(0) & \textit{if } \widetilde{W}_{j,k,t} = 0 \end{cases} \qquad Y_t^{0,obs} = \begin{cases} Y_{j,t}(0) & \textit{if } \widetilde{W}_{j,k,t} = w_{j,k} \\ Y_{j,t}(0) & \textit{if } \widetilde{W}_{j,k,t} = 0 \end{cases}$$

and the non observed outcome of the treated units under no treatment are

$$Y_t^{1,nobs} = \begin{cases} Y_{j,t}(0) & \textit{if } \widetilde{W}_{j,k,t} = w_{j,k} \\ Y_{j,t}(w_{j,k}) & \textit{if } \widetilde{W}_{j,k,t} = 0 \end{cases}$$

Assumption 16. *(Strong Parallel Trends). For each $j \geq 1$, $k \geq 1$,$t \geq 1$,*

$$\mathbb{E}[Y_{j,t}^1(w_{j,k}) - Y_{j,t}^0(0)|\widetilde{W}_{j,k,t} = w_{j,k}] = \mathbb{E}[Y_{j,t}^1(w_{j,k}) - Y_{j,t}^0(0)].$$

Assumption 15 in this case indicates that we observe the treatment assignment of treated units at each time, but we are incapable of observing them under a no treatment. The underlying assumption is that the control units have identical potential outcomes to what would be the one of the treated under no treatment. Assumption 16 is the same as the one of strong parallel trends by **?** but must be read differently. In this case, it indicates that the control unit eliminates the time selection bias. This lead to the following theorem:

Theorem 6. *(Identification of ACR in the direct control case). For each $j \geq 1$, $k \geq 1$,$t \geq 1$, if the policy variable is normally distributed and assumptions 15, 12, and 16, are satisfied, the Cholesky decomposition of the direct control approach to the CVAR identifies*

$$\gamma_{j,k} = \frac{\delta \mathbb{E}[Y_j(w_{j,k}) - Y_j(0)]}{\delta w_{j,k}} + \Delta_{AR} = ACR_{j,k}(Y_j|w_{j,k}, 0) + \Delta_{AR},$$

where Δ_{AR} is akin to Theorem 5.

This result should be read exactly like the previous one for the dummy policy variable case. Even in the case of a continuous policy variable, it is possible to obtain the ACR without relying on independence assumptions.

Section 4.2 will present an alternative to this set of assumptions with similar benefits to the dummy case.

4.2 Estimation through VECM

The C-VAR I propose is the following:

$$X_t = (W'_{1,t}, .., W'_{j,k,t}, Y^{1\prime}_{1,t}, .., Y^{1\prime}_{J,t}, Y^{0\prime}_{1,t}, .., Y^{0\prime}_{J,t})'.$$

The VECM estimates the long run relationship between variables through $\Pi = \alpha\beta'$, so that $X_t \sim \mathcal{I}(1)$ but $X_t - \beta X_t \sim \mathcal{I}(0)$. Traditionally, the model is written as follows

$$\Delta X_t = \Pi X_{t-1} + \sum_{l=1}^{p} A_l \Delta X_{t-l} + \epsilon_t, \tag{4}$$

where ΔX_t represents the first difference of the non-stationary variables. For such estimation to be credible, the following assumption needs to hold.

Assumption 17. *(The roots of the characteristic polynomial lie outside the unit circle). The roots of the characteristic polynomial*

$$\Pi(z) = (1 - z)I_p - \alpha\beta' z - \sum_{l=1}^{p-1} A_l (1 - z) z^l,$$

are outside the complex unit circle or at 1. Moreover, the matrices α and β have full column rank r. Further, define $\Psi = I_{k+(j\times 2)} - \sum_{l=1}^{p-1} A_l$ and assume full rank of the matrix

$$\alpha'_\perp \Psi \beta_\perp.$$

Hence, Theorem 4.2 of **?** holds as follows:

Theorem 7. *(Granger's Respresentation Theorem). Suppose assumption 17 is satisfied. Then, there exists a matrix $\Pi = \alpha\beta'$. Moreover, a necessary and sufficient condition that $\Delta X_t - \mathbb{E}[\Delta X_t]$ and $\beta' X_t - \mathbb{E}[\beta' X_t]$ can be given initial distributions such that they become $I(0)$ is that*

$$| - \alpha'_\perp A(1)\beta_\perp| = |\alpha'_\perp \Psi \beta_\perp| \neq 0, \tag{5}$$

In this case the solution of the VECM in Equation 4 can be written with the following representation

$$X_t = C \sum_{l=1}^{t} \epsilon_l + C_1(L)\epsilon_t + A, \tag{6}$$

where A depends on the initial values, such that $\beta' A = 0$, and where $C = \beta_\perp(\alpha'_\perp \Gamma \beta_\perp)^{-1}\alpha'_\perp$. It follows that X_t is a cointegrated $I(1)$ process with cointegrating vectors β. The function

$C_1(z)$ *satisfies*

$$A^{-1}(z) = C\frac{1}{1-z} + C_1(z), \quad z \neq 1, \tag{7}$$

where the power series for $C(z)$ *is convergent for* $|z| < 1 + \delta$ *for some* $\delta > 0$.

This theorem implies that any deviation from the cointegrating relationship is temporary and the processes inside the vector X_t revert to their common equilibrium. This implies that $\epsilon_t = \Delta X_t - \Pi X_{t-1} + \sum_{l=1}^{p} A_l \Delta X_{t-l}$ has a solution for each t.

Notice that such representation has a S-VECM equivalent representation that also includes the causal effects of interest, which are defined as A_0. Therefore, the structural error correction model equivalent (S-VECM) of the VECM representation in Equation 4 would be:

$$A_0 \Delta X_t = \alpha^* \beta' X_{t-1} + \sum_{l=1}^{p} A_l^* \Delta X_{t-l} + \epsilon_t^*, \quad \epsilon_t^* \sim \mathcal{D}_p^*(0, \Omega^*) \tag{8}$$

where $\alpha^* = A_0 \alpha$, $A_l^* = A_0 A_l$, $\mathcal{D}$ is a generic distribution (not necessarily normal), for all i, $\epsilon_t^* = A_0 \epsilon_t$ and $\Omega^* = A_0^* \Omega A_0^{*\prime}$. According to **?**, inference can be carried out as usual and the rank conditions apply, as well as the formulations in connection with the identification of β. The conclusion of this is that the presence of non-stationary variables allows two distinct identification problems to be formulated. First, the long-run relations must be identified uniquely in order to estimate and interpret them and then the short-run parameters $(\alpha^*, A_0^*, .., A_{k-1}^*)$ must be identified uniquely in the usual way.

4.2.1 Dummy policy in the VECM case

For the Choleksy decomposition to identify ATT, the trend will be assumed to be a credible counterfactual, in that the difference between the treated variable and the trend would be zero if the treated variable was to not be treated.

Assumption 18. *(Deviations from trend in treated times are causal): For each* $j \geq 1$,

$k \geq 1$,$t \geq 1$,

$$\mathbb{E}[\Delta Y_{j,t}^{1}(0)|\widetilde{W}_{j,k,t}=1] = \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^{p} A_l \Delta X_{t-l})|\widetilde{W}_{j,k,t}=1].$$

Assumption 19. *(Deviations from trend in non-treated times are zero): For each $j \geq 1$, $k \geq 1$,$t \geq 1$,*

$$\mathbb{E}[\Delta Y_{j,t}^{1}(0)|\widetilde{W}_{j,k,t}=0] = \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^{p} A_l \Delta X_{t-l})|\widetilde{W}_{j,k,t}=0].$$

Here $K+j$ indicates the row specific to the outcome variable of the treated country or unit. Notice that assumption 18 involves an untestable hypothesis since it involves counterfactuals assignment.[11] However, since it involves some estimated parameters, it only requires that a combination of Π and A_l that satisfies assumption 19 exists. That is because Theorem 7, implies that $\hat{\Pi}$ and $\hat{A}_l$ are specifically chosen to minimize the difference in the assumption 19. The asymptotics can be found in Ch. 13.4 of **?**.

Remark 6. Notice that assumptions 18 and 19 have an interpretation under the potential outcome framework in 11. In particular, define $\pi_{K+j,k+j*2}$ as the coefficient within Π that relates $\Delta Y_{j,t}^{1}$ and $Y_{j,t-1}^{0}$, and $f_{K+j}(\bar{\Pi}\bar{X}_{t-1})$ as the sum of the remaining coefficients. From the perspective of Rubin's causal model, assumption 18 is equivalent to the statement that, in treated times, the counterfactual is partially generated using a linear transformation of the realized outcomes of the lag of the controls. For example, consider the trivariate system $X_t = (W_t', Y_t^1, Y_t^0)'$ where Y_t^0 serves as a counterfactual trend for Y_t^1. Suppose the long-run behaviour is characterized by a single cointegrating relation that involves only Y_t^1 and Y_t^0. Using the standard factorization $\Pi = \alpha\beta'$ with

$$\beta = (0,\, 1,\, -\lambda)' \quad \text{and} \quad \alpha = (0,\, \alpha_2,\, 0)',$$

[11] This is a property shared by many causal models. For example, Regression Discountinuity designs and DiD designs exploit respectively units that are right before the cutoff and other unit's realized outcomes as counterfactuals.

we obtain

$$\Pi = \alpha\beta' = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \alpha_2 & -\alpha_2\lambda \\ 0 & * & * \end{pmatrix}.$$

Writing $c_1 = \alpha_2$ and $c_0 = -\alpha_2\lambda$, the long-run part of the Y^1-equation can be written (ignoring short-run dynamics and shocks for expositional clarity) as

$$\Delta Y_t^1(0) = c_1 Y_{t-1}^1(0) + c_0 Y_{t-1}^0(0). \tag{9}$$

From such alternative definition it is possible to see that there is a potential outcome representation of the controls, but that is different from the traditional causal inference for two reasons. First, the counterfactual (right hand side of equation 9) is essentially lagged. Second, the counterfactual is a linear transformation (through c_1 and c_2) of the realized outcomes of the control and treated variables.

An advantage of cointegration is that it avoids the potential effect of violations of the underlying no spillover assumptions. This is because generating the counterfactual using lagged values of the controls avoids the potential invalidating effect of spillovers altogether.

Under such restrictions, the following theorem holds true.

Theorem 8. *(Identification of ATT in the VECM case). For each $j \geq 1$, $k \geq 1$,$t \geq 1$, if the policy variable is a dummy and assumptions 18 and 19 are satisfied, the Cholesky decomposition estimated with the VECM approach to the CVAR identifies*

$$\gamma_{j,k} = \mathbb{E}[\Delta Y_{j,t}(1) - \Delta Y_{j,t}(0)|\widetilde{W}_{j,k,t} = 1] = ATT_{j,k}(\Delta Y_j|1,0|1).$$

4.2.2 Continuous policy in the VECM case

For the Choleksy decomposition to identify the ACR in the normally distributed policy case, the trend will be assumed to be a credible counterfactual, in that the difference between the treated variable and the trend would be zero if the treated variable was to not be treated by any amount $w_{j,k}$.

Assumption 20. *(Continuous deviations from trend in treated times are causal): For each $j \geq 1$, $k \geq 1$,$t \geq 1$,*

$$\mathbb{E}[\Delta Y_{j,t}^{1}(0)|\widetilde{W}_{j,k,t} = w_{j,k}] = \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^{p} A_l \Delta X_{t-l})|\widetilde{W}_{j,k,t} = w_{j,k}].$$

Assumption 21. *(Continuous strong parallel trends): For each $j \geq 1$, $k \geq 1$, $t \geq 1$,*

$$\mathbb{E}[\Delta Y_{j,t}^{1}(w_{j,k}) - \Delta Y_{j,t}^{1}(0)|\widetilde{W}_{j,k,t} = w_{j,k}] = \mathbb{E}[\Delta Y_{j,t}^{1}(w_{j,k}) - \Delta Y_{j,t}^{1}(0)]$$

Assumption 20 is required to define the counterfactual generated from the control unit to be credible at any time, a condition similar to assumption 18. Assumption 21 instead is used to eliminate the selection bias component, a condition which credibility depends on the endogeneity of the residuals of the policy.

This restriction is sufficient to lead to the following theorem

Theorem 9. *(Identification of ACR in the VECM case). For each $j \geq 1$, $k \geq 1$,$t \geq 1$, if the policy variable is normally distributed and assumptions 20 and 21 are satisfied, the Cholesky decomposition of the VECM of the CVAR identifies*

$$\gamma_{j,k} = \frac{\delta \mathbb{E}[\Delta(Y_j^1(w_{j,k}) - \Delta Y_j^1(0)]}{\delta w_{j,k}} = ACR_{j,k}(\Delta Y_j | w_{j,k}, 0).$$

One potential issue for macroeconomists would be the need to restrict the control series to a conservative set. This is because adding all the available controls to the VAR systematically increases the number of coefficients to be estimated. Therefore, the applied researcher should be careful about finding an appropriate set of series to be used as counterfactuals. Many possible criteria could be used for such minimization. In the following paragraph I will simply order the candidate controls according to the Likelihood Ratio test of the resulting VECM using different sets of controls. Another approach to the problem could be using many different counterfactuals when the time component is large enough.

5 The effects of natural disasters on Industrial Production and Macroeconomic Uncertainty

? (henceforth LMN) conducted an estimation of an S-VAR model with 6 lags encompassing the Cost of natural Disasters (CD), Industrial Production (IP), and Macroeconomic Uncertainty (MU) index proposed by **?**. I plot the original variables and the German counterpart in Figure 1. From the figure, several relevant features emerge. The series suggest a certain degree of contemporaneous comovements, which are especially relevant in moments in which the variables appear to change during the great financial crisis. Therefore, even outside the scope of using Germany as a counterfactual for inferential purposes, adding such data to the system may provide many modeling advantages: it may take care of common breaks, common structural factors, and many different elements that perturbate the two economies. In the original application, however, the focus is only on US data. There, the causal assumption is that the cost of natural disaster is considered exogenous. Such an assumption allows the authors to employ a Cholesky decomposition on a VAR composed by the CD first, followed by IP and MU. Their objective is to simulate a shock equivalent to the magnitude of the COVID-19 pandemic and utilize the simulation to predict the evolution of IP and MU.

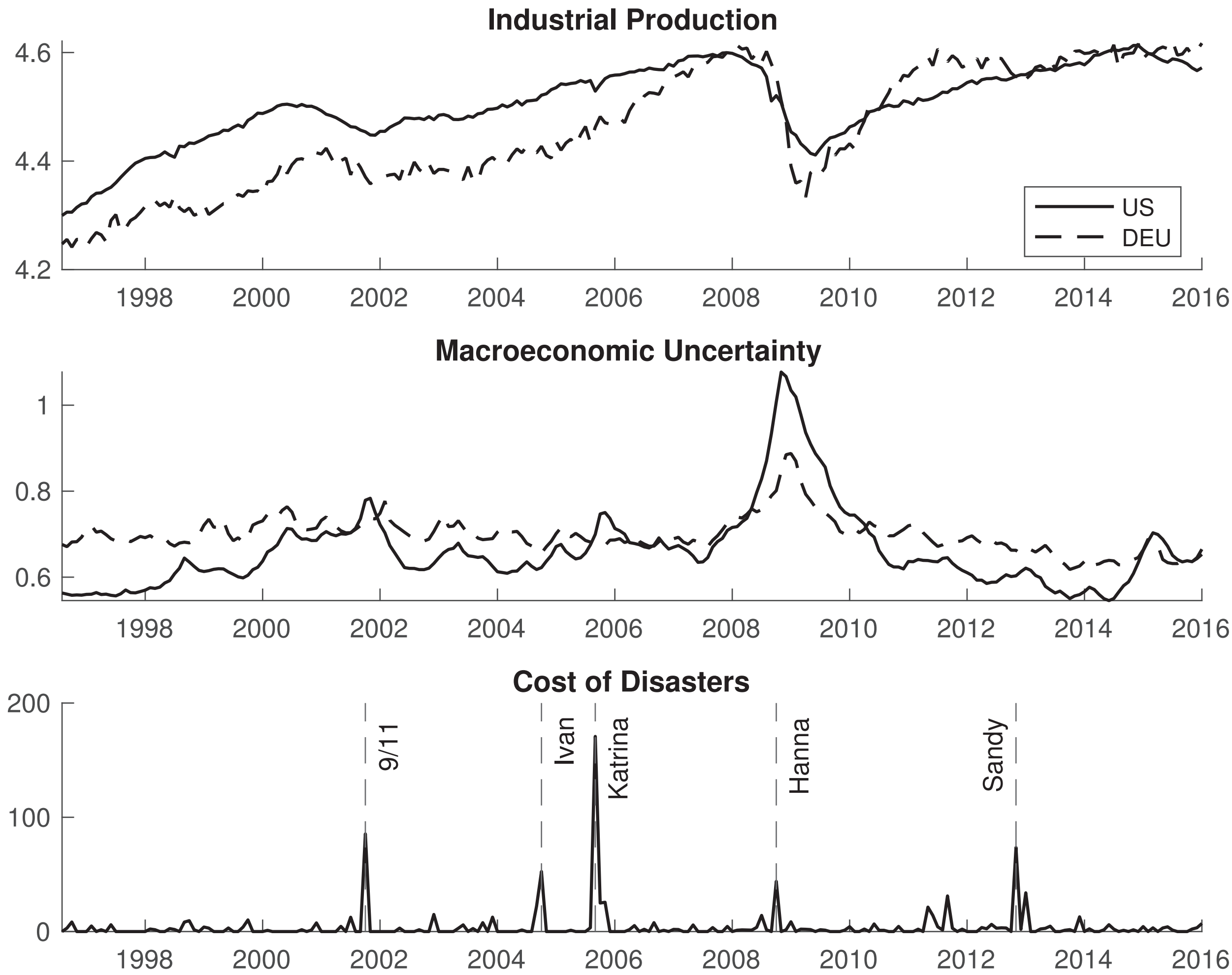

Figure 1: Industrial Production, Macroeconomic Uncertainty and Cost of Natural Disasters in the United States and Germany.

Although not explicitly mentioned in the paper, it is worth noting that the selected outcome variables exhibit non-stationarity. To address this, LMN apply the detrendization method proposed by **?** to filter out the trend component. While such an approach is not necessarily problematic in the view of the most classical identification in macroeconomics literature, it presents two challenges. The first one relates to the interpretability of the model. Since the CD variable is continuous and non-negative, its effect will be, as discussed in Theorem 4, a weighted average of a filtered ATE and ACRT. Therefore, the estimated Impulse Response Function is capturing both the deviation with respect to zero, and the deviation with respect to progressively different values. Intuitively, the pitfall of the identification is represented by the high discontinuity of the causing variable. The second one relates to the likelihood of the exogeneity assumption to hold. In fact

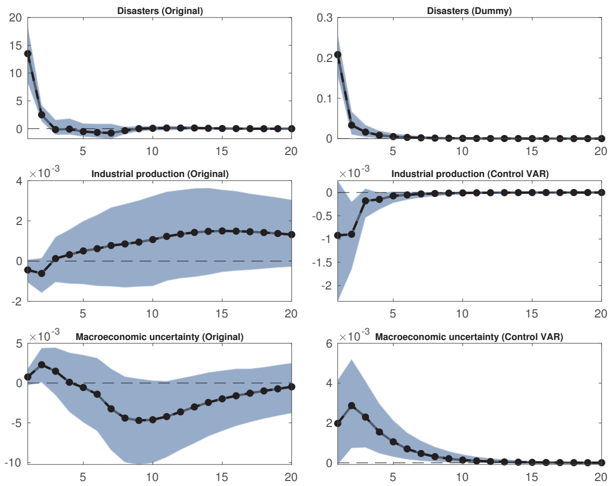

Figure 2: Impulse Response Functions of the filtered variables to a 1% shock in the CD. Confidence intervals are bootstrapped at the 95% level using wild bootstrap. The left panel are obtained through a simple VAR with filtered US data. The right panels result from a VECM estimated using also German data, where the disasters are transformed to a dummy indicating the 95-th quantile of the disaster variable.

the authors notice that the most relevant events are the Heat Wave of 1980, September 11, Hurricane Katrina, Hurricane Maria, and Hurricane Harvey. The fact that all those episodes happened in August and September may contaminate inference due to underlying unobservables. Many other instances could invalidate inference when the underlying assumption is the complete randomness of the natural disaster series, which may motivate approaches to causal inference that do not rely on independence.

The three left panels of figure 2 plot the original results for horizons extended to 20 periods.

The effects of the CD on IP are negative for the first two periods and positive in the following ones. The effects of the CD on MU are positive for three periods and negative afterward. Those responses may indicate the filtering procedure's ineffectiveness in changing the impulse response's shape. [12]

To solve the first issue (the interpretability of the mix of filtered effects of a non-negative continuous variable), an alternative to such procedure could be using a dummy variable that is equal to 1 when a particularly disruptive disaster happens and 0 otherwise, with the goal of identifying the effect of a high cost disaster happening. This other approach

[12]Subsection A.0.2 provides some evidence in support of the fact that the Impulse Responses are not affected by the filtering.

could move the interpretation of the causal effects from the ones of a non-negative policy to the ones of a dummy policy. [13]. Until now, VARs did not have the means to solve the second issue (exogeneity) within a proper causal framework that declaratively assigns a counterfactual.

My alternative modeling approach consists of two deviations from the original model. The first one is using a dummy that represents when the Cost of Natural Disasters (CD) are greater than the 95-*th* quantile of the time series -or above a cost of 150 million- in place of the CD series itself. The second one is using a control for the United States data. I use the MU data from **?**, which collects the same type of variable as the work of **?**. Out of the countries available in their dataset, Germany appears to be the one that provides the best-identified system according to the battery of tests discussed in Appendix A.0.1.[14]

Accordingly, I estimate a VECM for the following system:

$$X_t = (CD^{95\prime}_{US,t}, IP'_{US,t}, MU'_{US,t}, IP'_{DEU,t}, MU'_{DEU,t})',$$

where $CD^{95\prime}_{US,t}$ represents the 95-*th* quantile of the CD series, $IP'_{US,t}$ represents IP in the US, $MU'_{US,t}$ represents MU in the US, $IP'_{DEU,t}$ represents IP in Germany, $MU'_{DEU,t}$ represents MU in Germany. This method relies on assumptions 18 and 19 instead of the independence of the policy and captures the effect of a natural disaster of high proportion versus a null or not costly disaster.

The resulting Impulse Response Functions are displayed in the right-hand side of Figure 2. Two particular features emerge from the picture. The first one is that the horizon of the impact of a natural disaster shock on IP is short-lived and always negative. This result is particularly relevant for policymakers, since it clarifies the ambiguous role that natural disasters play in the economy. A similar feature emerges from the response of MU to the disasters dummy, but it should be noted that uncertainty seems to exhibit

[13] Appendix A.0.4 provides evidence in support of the fact that such modification may not substantially change the IRF. In that case, I use a VAR model where the first variable is a dummy that is equal to 1 in periods in which the CD variable is above its 95-*th* quantile.

[14] I also source data on the cost of natural disasters in Germany from the International Disasters database to control for the possible presence of natural disasters in Germany. The results are invariant to the addition of a dummy variable that controls for Germany disasters.

persistence but no cyclicality. The second one is that the confidence intervals shrink to zero together with the causal responses.

Both the results come from a particular feature of my alternative model. Adding another source of information (German data) results in a preferred system that only includes one lag, which makes the response of the outcome variables to a disaster shock shorter. Short lived impulse responses are a property of VARs with low lags and AR(1) coefficients far from the unit root.

The overall conclusion is that such an approach to modeling presents two different advantages. The first one is related to the causal interpretability of the model. An ATT may be more interpretable than an ACRT+ATE, and may be preferred over an ATE because no independence assumption would be required. The second is related to a pure macroeconomic interpretation. A property of VECM models is that they tend to reject lags higher than one, which may be an advantage if the objective is the interpretation of the short-run effects of a policy.

In appendix A I discuss more details about the empirical application, the lag choice, and discuss alternative specifications with many different controls.

6 Conclusions

Traditional VAR estimators may have a broader range of causal interpretations than the ones discussed by **?**. I show that VAR may identify other types of causal effects. In the case of normal distribution of the policy innovations, the effect identified becomes an ACR under their same independence assumptions. In the case of a non-negative distribution of the policy the effect identified is the sum of an ACRT and an ATE.

If controls are available, cointegration can be exploited, and the resulting coefficient may still have a causal interpretation. In the case of a dummy policy variable, they can identify an ATT. In the case of a continuous and normal distribution, they identify an ACR. This approach provides two substantial advantages. First, it relaxes potentially hard to satisfy independence assumptions. Second, it may deliver more credible impulse responses.

I discuss all of these advantages in the context of the work of **?**, which analyzes the causal effects of natural disasters on the US economy.

A Additional results for the empirical application

A.0.1 Additional results for the selection of Germany as a control

Table 1 represents multiple LR tests ran on different VAR specifications that include the disaster dummy, IP and MU in the US and IP an MU in different European countries. For each of the nations I run a VAR on

$$X_t^{control} = (CD_{US,t}^{95\prime}, IP_{US,t}^{\prime}, MU_{US,t}^{\prime}, IP_{control,t}^{\prime}, MU_{control,t}^{\prime})^{\prime}$$

and let *control* vary according to what Nation is utilized as a control. This approach has been used in the C-VAR and common feature literature (see, among others **?**). I select Germany since it outperforms the others for $r = 3$.

LR	Germany	Italy	France	Spain	Critical
r=1	**206.15**	**205,76**	**204.39**	**217.43**	97.18
r=2	**127.29**	**125.97**	**119.61**	**126.85**	71.88
r=3	**54.90**	48.75	45.25	43.56	49.65
r=4	20.09	21.58	10.98	14.23	32.00
r=5	4.29	0.22	1.62	0.97	17.85
r=6	0.00	0.00	0.00	0.00	7.52

Table 1: LR test of 5 variable VARs.

After determining the optimal control, the next steps for the VAR identification regard the choice of lags. Table 2 collects the BIC tests for each different lag of the VECM. The lag 1 is preferred over the other specifications.

p	BIC
1	-6347,94
2	-6261,51
3	-6157,23
4	-6036,27
5	-5918,55
6	-5786,35

Table 2: BIC test for each lag p in the case of a model with just Germany as control.

? discuss the reasons why a Breush-Godfrey type test could be optimal for the case of unknown cointegration rank and unknown lags[15]. Table 3 shows that using just one lag results in a well specified model for which the null of the absence of residual heteroskedasticity cannot be rejected at any confidence level. The critical values, reported in column 3 to 5 have been extrapolated from a $\chi^2(K+J)$ distribution.

	Breush -Godfrey test	$\alpha = .90$	$\alpha = .95$	$\alpha = .99$
$p = 1$	0.8	9.24	11.07	15.09
$p = 2$	0.95	15.99	18.31	23.21
$p = 3$	1.11	22.31	25	30.58
$p = 4$	1.26	28.41	31.41	37.57
$p = 5$	1.41	34.38	37.65	44.31
$p = 6$	1.48	40.26	43.77	50.89

Table 3: Breush-Godfrey test results and critical values in the case of just Germany as control.

A.0.2 Non-filtered series

Figure 3 shows the results of a VAR estimated using the raw series (non-stationary). While in violation of the Wold theorem, the Impulse Responses appear to be immune to

[15]The Breush-Godfrey is calculated on a VECM model estimated through generalised least squares as suggested by the authors.

the filtering of the series. This may be interpreted as indirect evidence of the invulnerability of the IRF to the filtering proposed by **?**.

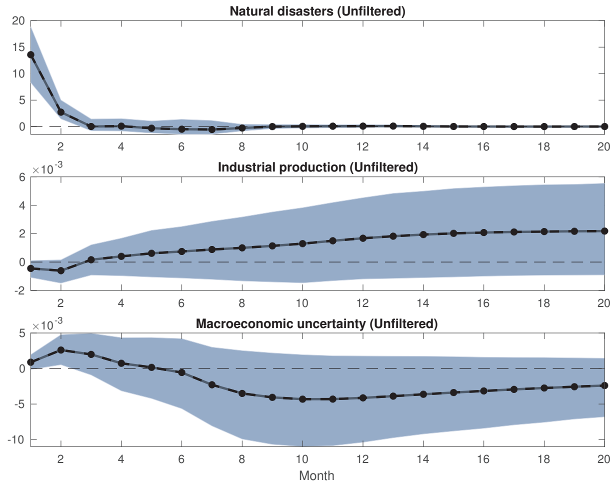

Figure 3: Impulse Response Functions of the non filtered variables to a 1% shock in the disasters dummy. Confidence intervals are bootstrapped at the 95% level using wild bootstrap.

A.0.3 Alternative with many controls

I consider the alternative model that includes all the possible controls, so that the VAR becomes a high dimensional model of the kind:

$$X_t^{big} = (CD^{95\prime}_{US,t}, IP'_{US,t} MU'_{US,t}, , IP'_{DEU,t}, IP'_{ITA,t}, IP'_{FRA,t}, IP'_{ESP,t}, MU'_{DEU,t}, MU'_{ITA,t}, MU'_{FRA,t}, MU'_{ES}$$

The BIC test results, presented in Table 4, indicate that the lag 1 model is preferred.

p	BIC
1	-14129.55
2	-13599.25
3	-13125.91
4	-12564.09
5	-12024.62
6	-11556.25

Table 4: BIC test for each lag p in the case of many controls.

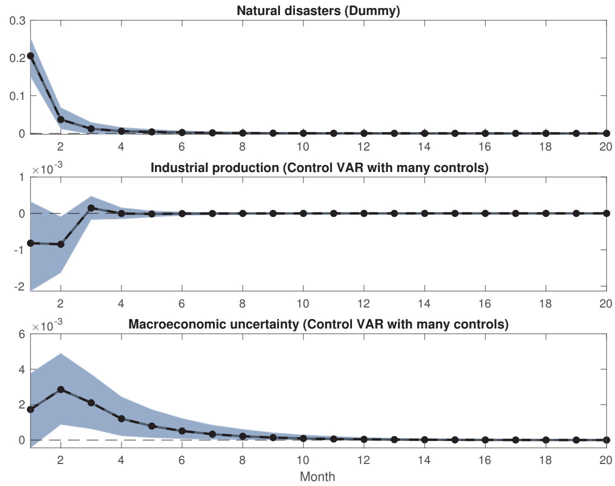

Figure 4: Impulse Response Functions to a 1% shock in the disasters dummy in the case of many controls. Confidence intervals are bootstrapped at the 95% level using wild bootstrap.

Similar conclusions can be drawn from the Breush-Godfrey test in Table 5.

	Breush Godfrey test	$\alpha = .90$	$\alpha = .95$	$\alpha = .99$
$p = 1$	1.42	9.24	11.07	15.09
$p = 2$	1.9	15.99	18.31	23.21
$p = 3$	2.49	22.31	25	30.58
$p = 4$	2.99	28.41	31.41	37.57
$p = 5$	3.49	34.38	37.65	44.31
$p = 6$	4.07	40.26	43.77	50.89

Table 5: Breush Godfrey test results and critical values in the case of many controls.

Finally, Figure 4 represents the response of Industrial Production and Macroeconomic Uncertainty to a natural disaster shock dummy in the case of multiple controls.

A.0.4 Dummy variable in the case of no controls

Figure 5 represents the Impulse Response of the system

$$X_t^{95,or} = (CD_{US,t}^{95\prime}, IP_{US,t}^{\prime}, MU_{US,t}^{\prime})^{\prime}.$$

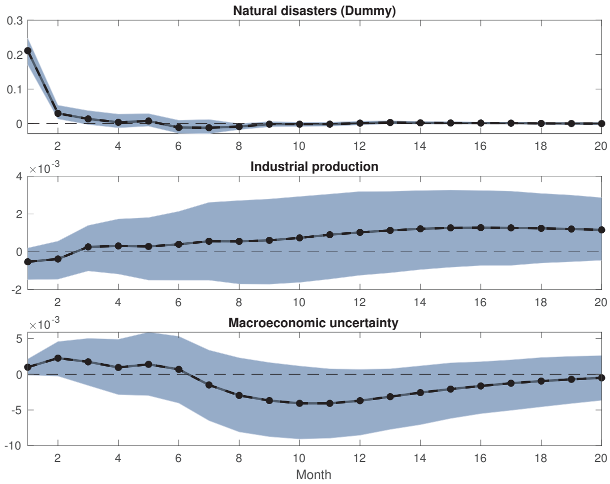

Figure 5: Impulse Response Functions to a 1% shock in the disasters dummy in a VAR with no controls. Confidence intervals are bootstrapped at the 95% level using wild bootstrap.

It indicates that the response to a natural disaster shock is similar to the case of a continuous variable. It provides evidence of the fact that the IRFs generated through the VECM are different because of cointegration, and not because of the change of the natural disasters variable to a dummy. Such figure can be interpreted as the identification of a filtered ATE under exogeneity conditions a-là-**?**.

B Mathematical results

Proof. **Proof of Theorem 1.** The proof is akin to the one of Theorem 1 in **?**, substituting $\widetilde{Y}_{j,t}$ for $Y_{j,t}$. $\square$

Proof. **Proof of Theorem 2.** Consider that the estimator is $\gamma_{j,k} = Cov(\widetilde{Y}_j, \widetilde{W}_{j,k})/var(\widetilde{W}_{j,k})$. Where the numerator is

$$
\begin{aligned}
Cov(\widetilde{Y}_j, \widetilde{W}_{j,k}) &= \mathbb{E}[(\widetilde{Y}_j - \mathbb{E}(\widetilde{Y}_j))(\widetilde{W}_{j,k} - \mathbb{E}(\widetilde{W}_{j,k}))] \\
&= \mathbb{E}[\widetilde{Y}_j(\widetilde{W}_{j,k} - \mathbb{E}(\widetilde{W}_{j,k}))] \\
&= \mathbb{E}[\widetilde{W}_{j,k} - \mathbb{E}(\widetilde{W}_{j,k})]\mathbb{E}[\widetilde{Y}_j|\widetilde{W}_{j,k} = w_{j,k}] \\
&= \int (w_{j,k} - \mathbb{E}(\widetilde{W}_{j,k}))g_j(w_{j,k})f_{\widetilde{W}_{j,k}}(w_{j,k})dw_{j,k}
\end{aligned}
$$

Where the first equality holds because of the law of the covariance, the second holds because the innovations of a VAR are assumed to be zero mean for each j and each t according to assumption 8, the third equality rewrites $\widetilde{Y}_j$ in terms of the realized values of $\widetilde{W}_{j,k}$, defined as $w_{j,k}$, and the last holds by rewriting the expected value as an integral and defining $g_j(w_{j,k}) = \mathbb{E}[\widetilde{Y}_j|\widetilde{W}_{j,k} = w_{j,k}]$. Defining $v'(m) = (w_{j,k} - \mathbb{E}[\widetilde{W}_{j,k}])f_{\widetilde{W}_{j,k}}(m)$, $v(m) = \int_{-\infty}^{\widetilde{W}_{j,k}}(m - \mathbb{E}[\widetilde{W}_{j,k}])f_{\widetilde{W}_{j,k}}(m)dm$ and $u_j(w_{j,k}) = g_j(w_{j,k})$ I can apply integration by parts to obtain

$$
\begin{aligned}
\mathrm{Cov}(\widetilde{Y}_j, \widetilde{W}_{j,k}) = &\int_{-\infty}^{\widetilde{W}_{j,k}} (m - \mathbb{E}[\widetilde{W}_{j,k}])f_{\widetilde{W}_{j,k}}(m)\,dm\, g_j(w_{j,k}) \\
&- \int_{-\infty}^{\infty} \Big(\int_{-\infty}^{\widetilde{W}_{j,k}} (m - \mathbb{E}[\widetilde{W}_{j,k}])f_{\widetilde{W}_{j,k}}(m)\,dm \Big) g_j'(w_{j,k})\,dw_{j,k},
\end{aligned} \tag{10}
$$

notice that the first part converges to zero if the variance of $\widetilde{W}_{j,k}$ exists, and we obtain

$$
\begin{aligned}
Cov(\widetilde{Y}_j, \widetilde{W}_{j,k}) &= \int_{-\infty}^{\infty} (\int_{-\infty}^{\widetilde{W}_{j,k}} (\mathbb{E}[\widetilde{W}_{j,k}] - m)f_{\widetilde{W}_{j,k}}(m)dm\, g_j'(w_{j,k})dw_{j,k} \\
&= \int_{-\infty}^{\infty} (\mathbb{E}[\widetilde{W}_{j,k}]F_{\widetilde{W}_{j,k}}(w_{j,k}) - \theta_{\widetilde{W}_{j,k}}(w_{j,k}))g_j'(w_{j,k})dw_{j,k}.
\end{aligned}
$$

Where the first equality holds by changing the sign of the second part of 10, the second holds by substituting the definition of $\theta_{\widetilde{W}_{j,k}}(w_{j,k}) = \int_{-\infty}^{\widetilde{W}_{j,k}} m f_{\widetilde{W}_{j,k}}(m)dm$. And the denominator is $var(\widetilde{W}_{j,k}) = \sigma^2_{\widetilde{W}_{j,k}}$. Hence the estimated coefficient becomes

$$\gamma_{j,k} = \frac{\int_{-\infty}^{\infty}(\mathbb{E}[\widetilde{W}_{j,k}]F_{\widetilde{W}_{j,k}}(w_{j,k}) - \theta_{\widetilde{W}_{j,k}}(w_{j,k}))g_j'(w_{j,k})dw_{j,k}}{\sigma^2_{\widetilde{W}_{j,k}}}.$$

which is equivalent to the one in the theorem by using the definition of the weights $q(w_{j,k}) = \frac{1}{\sigma^2_{\widetilde{W}_{j,k}}}\int_{-\infty}^{\infty}(\mathbb{E}[\widetilde{W}_{j,k}]F_{\widetilde{W}_{j,k}}(w_{j,k}) - \theta_{\widetilde{W}_{j,k}}(w_{j,k}))$. I can rewrite the definition of $\gamma_{j,k}$ according to a set of weights $q(w_{j,k})$ (notice that they do not depend on the outcome variable) and the function $g_j'(w_{j,k})$:

$$\gamma_{j,k} = \int q(w_{j,k})g_j'(w_{j,k})dw_{j,k},$$

Substituting $q(w_{j,k}) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{w_{j,k}} me^{-m^2/2}dm = \frac{1}{\sqrt{2\pi}}e^{-m^2/2}$ inside the definition of γ_{jk} I obtain the following

$$\begin{aligned}
\gamma_{j,k} &= \int \frac{1}{\sqrt{2\pi}}e^{-m^2/2}g_j'(w_{j,k})dw_{j,k} \\
&= g_j'(w_{j,k})\int \frac{1}{\sqrt{2\pi}}e^{-m^2/2}dw_{j,k} \\
&= g_j'(w_{j,k})
\end{aligned}$$

Where the first equality is true by substituting the value of the weights $q(w_{j,k})$, the second by the fact that the density of $g_j'(w_{j,k})$ does not depend on $w_{j,k}$, and the last by the laws of integration of normal variables which are satisfied by $\widetilde{W}_{j,k}$ according to assumption 8. □

Proof. (**Proof of Theorem 3**). Notice that the ACRT is equivalent to a

$$\frac{\delta\mathbb{E}[\widetilde{Y}_j(w_{j,k})|\widetilde{W}_{j,k} = w_{j,k}]}{\delta w_{j,k}} = \frac{\delta\mathbb{E}[\widetilde{Y}_j(w_{j,k})]}{\delta w_{j,k}} + \frac{cov(\widetilde{Y}_j(w_{j,k}), 1\{\widetilde{W}_{j,k} = w_{j,k}\})}{var(1\{\widetilde{W}_{j,k} = w_{j,k}\})}.$$

By assumption 7

$$cov(\widetilde{Y}_j(w_{j,k}), 1\{\widetilde{W}_{j,k} = w_{j,k}\}) = 0$$

would be implied by

$$\widetilde{W}_{j,k} \perp \widetilde{Y}_j(w_{j,k})$$

which is implied by

$$\widetilde{W}_{j,k} \perp \{\widetilde{Y}_j(w_{j,k}) : w_{j,k} \in \mathcal{W}_k\}.$$

This concludes the proof. $\square$

Proof. **(Proof of Theorem 4).** The proposed estimator is: $\gamma_{j,k} = cov(\widetilde{Y}_j, \widetilde{W}_{j,k})/var(\widetilde{W}_{j,k})$. To shorten the definitions I will use $m(d) = \mathbb{E}[\widetilde{Y}_j|\widetilde{W}_{j,k} = d]$.

$$\begin{aligned}
\gamma_{j,k} =& \frac{\mathbb{E}[\widetilde{Y}_j(\widetilde{W}_{j,k} - \mathbb{E}[\widetilde{W}_{j,k}])]}{var(\widetilde{W}_{j,k})} \\
&= \mathbb{E}[\frac{(\widetilde{W}_{j,k} - \mathbb{E}[\widetilde{W}_{j,k}])}{var(\widetilde{W}_{j,k})}(m(\widetilde{W}_{j,k}) - m(0))] \\
&= \mathbb{E}\left[\frac{(\widetilde{W}_{j,k} - \mathbb{E}[\widetilde{W}_{j,k}])}{var(\widetilde{W}_{j,k})}(m(\widetilde{W}_{j,k}) - m(0))|\widetilde{W}_{j,k} > 0\right]\mathbb{P}(\widetilde{W}_{j,k} > 0) \\
&= \mathbb{E}\left[\frac{(\widetilde{W}_{j,k} - \mathbb{E}[\widetilde{W}_{j,k}])}{var(\widetilde{W}_{j,k})}(m(\widetilde{W}_{j,k}) - m(d_L))|\widetilde{W}_{j,k} > 0\right]\mathbb{P}(\widetilde{W}_{j,k} > 0) + \\
&+ \mathbb{E}\left[\frac{(\widetilde{W}_{j,k} - \mathbb{E}[\widetilde{W}_{j,k}])}{var(\widetilde{W}_{j,k})}(m(d_L) - m(0))|\widetilde{W}_{j,k} > 0\right]\mathbb{P}(\widetilde{W}_{j,k} > 0) = \\
&= A_1 + A_2
\end{aligned}$$

where the first equality holds because the proposed estimator is akin to a simple linear regression of $\widetilde{Y}_j$ on $\widetilde{W}_{j,k}$, the second equality holds because $\mathbb{E}[(\widetilde{W}_{j,k} - \mathbb{E}[\widetilde{W}_{j,k}])m(0)] = 0$, the third equality holds because $\mathbb{E}[m(\widetilde{W}_{j,k}) - m(0)|\widetilde{W}_{j,k} = 0] = 0$, and the fourth equality holds by adding and subtracting $m(d_L)$. Then, for A_1:

$$
\begin{aligned}
A_1 =&\mathbb{E}\left[\frac{(\widetilde{W}_{j,k}-\mathbb{E}[\widetilde{W}_{j,k}])}{var(\widetilde{W}_{j,k})}(m(\widetilde{W}_{j,k})-m(d_L))|\widetilde{W}_{j,k}>0\right]\mathbb{P}(\widetilde{W}_{j,k}>0)\\
=&\frac{\mathbb{P}(\widetilde{W}_{j,k}>0)}{var(\widetilde{W}_{j,k})}\int_{d_L}^{d_U}(k-\mathbb{E}[\widetilde{W}_{j,k}])(m(k)-m(d_L))dF_{\widetilde{W}_{j,k}|\widetilde{W}_{j,k}>0}(k)\\
=&\frac{\mathbb{P}(\widetilde{W}_{j,k}>0)}{var(\widetilde{W}_{j,k})}\int_{d_L}^{d_U}(k-\mathbb{E}[\widetilde{W}_{j,k}])\int_{d_L}^{k}m'(w)dwdF_{\widetilde{W}_{j,k}|\widetilde{W}_{j,k}>0}(k)\\
=&\frac{\mathbb{P}(\widetilde{W}_{j,k}>0)}{var(\widetilde{W}_{j,k})}\int_{d_L}^{d_U}(k-\mathbb{E}[\widetilde{W}_{j,k}])\int_{d_L}^{d_U}\mathbf{1}\{w<k\}m'(w)dwdF_{\widetilde{W}_{j,k}|\widetilde{W}_{j,k}>0}(k)\\
=&\frac{\mathbb{P}(\widetilde{W}_{j,k}>0)}{var(\widetilde{W}_{j,k})}\int_{d_L}^{d_U}m'(w)\int_{d_L}^{d_U}(k-\mathbb{E}[\widetilde{W}_{j,k}])\mathbf{1}\{w<k\}dF_{\widetilde{W}_{j,k}|\widetilde{W}_{j,k}>0}(k)dw\\
=&\frac{\mathbb{P}(\widetilde{W}_{j,k}>0)}{var(\widetilde{W}_{j,k})}\int_{d_L}^{d_U}m'(w)\mathbb{E}[(\widetilde{W}_{j,k}-\mathbb{E}[\widetilde{W}_{j,k}])\mathbf{1}\{w\leq\widetilde{W}_{j,k}\}|\widetilde{W}_{j,k}>0]dw\\
=&\frac{\mathbb{P}(\widetilde{W}_{j,k}>0)}{var(\widetilde{W}_{j,k})}\int_{d_L}^{d_U}m'(w)\mathbb{E}[(\widetilde{W}_{j,k}-\mathbb{E}[\widetilde{W}_{j,k}])|\widetilde{W}_{j,k}\geq w]\mathbb{P}[\widetilde{W}_{j,k}\geq w|\widetilde{W}_{j,k}>0]dw\\
=&\int_{d_L}^{d_U}m'(w)\frac{(\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k}\geq w]-\mathbb{E}[\widetilde{W}_{j,k}])\mathbb{P}[\widetilde{W}_{j,k}\geq w]}{var(\widetilde{W}_{j,k})}dw
\end{aligned}
$$

where the first equality rewrites the previous result, the second holds by rearranging and writing the expectation as an integral, the third makes use of the fundamental theorem of calculus, the fourth rewrites the inner integral so that it is over d_U to d_L, the fifth changes the order of integration and rearranging terms, the sixth by rewriting the inner integral as an expectation, the seventh by the law of iterated expectations (and since $\widetilde{W}_{j,k}\geq w$ implies that $\widetilde{W}_{j,k}\geq 0$), and the last equality holds by combining terms. For A_2:

$$
\begin{aligned}
A_2 =& \mathbb{E}\left[\frac{(\widetilde{W}_{j,k}-\mathbb{E}[\widetilde{W}_{j,k}])}{var(\widetilde{W}_{j,k})}(m(d_L)-m(0))|\widetilde{W}_{j,k}>0\right]\mathbb{P}[\widetilde{W}_{j,k}>0]\\
=&\frac{(\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k}>0]-\mathbb{E}[\widetilde{W}_{j,k}])\mathbb{P}(\widetilde{W}_{j,k}>0)d_L}{var(\widetilde{W}_{j,k})}\frac{(m(d_L)-m(0)}{d_L}
\end{aligned}
$$

where the first equality is the definition of A_2, and the second holds by multiplying and dividing by d_L. Then:

$$\gamma_{j,k} = \int_{d_L}^{d_U} m'(w) \frac{(\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} \geq w] - \mathbb{E}[\widetilde{W}_{j,k}])\mathbb{P}[\widetilde{W}_{j,k} \geq w]}{var(\widetilde{W}_{j,k})} dw + \\ + \frac{(\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} > 0] - \mathbb{E}[\widetilde{W}_{j,k}])\mathbb{P}(\widetilde{W}_{j,k} > 0)d_L}{var(\widetilde{W}_{j,k})} \frac{(m(d_L) - m(0)}{d_L}$$

Which can be rewritten as:

$$\gamma_{j,k} = \int_{d_L}^{d_U} q_1(w) \frac{\delta \mathbb{E}[\widetilde{Y}_j|\widetilde{W}_{j,k} = w]}{dw} dw + q_0 \frac{\mathbb{E}[\widetilde{Y}_j|\widetilde{W}_{j,k} = d_L] - \mathbb{E}[\widetilde{Y}_j|\widetilde{W}_{j,k} = 0]}{d_L}$$

where:

$$q_1(w) := \frac{\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} \geq w] - \mathbb{E}[\widetilde{W}_{j,k}])\mathbb{P}(\widetilde{W}_{j,k} \geq w)}{var(\widetilde{W}_{j,k})} \text{ and } \quad q_0 := \frac{(\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} > 0] - \mathbb{E}[\widetilde{W}_{j,k}])\mathbb{P}(\widetilde{W}_{j,k} > 0)}{var(\widetilde{W}_{j,k})}$$

Where the weights satisfy:

$$\begin{aligned} \int_{d_L}^{d_U} q_1(w)dw + w_0 = & \frac{1}{var(\widetilde{W}_{j,k})} \{ \int_{d_L}^{d_U} \mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} \geq w] \mathbb{P}[\widetilde{W}_{j,k} \geq w)dw \\ & - \mathbb{E}[\widetilde{W}_{j,k}] \int_{d_L}^{d_L} \mathbb{P}(\widetilde{W}_{j,k} \geq w)dw \\ & + \mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} > 0]\mathbb{P}(\widetilde{W}_{j,k} > 0)d_L \\ & - \mathbb{E}[\widetilde{W}_{j,k}]\mathbb{P}(\widetilde{W}_{j,k} > 0)d_L \} \\ & := \frac{1}{var(\widetilde{W}_{j,k})} \{B_1 + B_2 + B_3 + B_4\} \end{aligned}$$

Where for B_1, for all $w \in \mathcal{W}_+$:

$$\begin{aligned} \mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} \geq w]\mathbb{P}(\widetilde{W}_{j,k} \geq w) &= \mathbb{E}[\widetilde{W}_{j,k}\mathbf{1}\{\widetilde{W}_{j,k} > w\}|\widetilde{W}_{j,k} \geq w]\mathbb{P}[\widetilde{W}_{j,k} \geq w] \\ &= \mathbb{E}[\widetilde{W}_{j,k}\mathbf{1}\{\widetilde{W}_{j,k} \geq w\}] \end{aligned}$$

which holds by the law of iterated expectations and implies that

$$
\begin{aligned}
B_1 &= \int_{d_L}^{d_U} \mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} \geq w]\mathbb{P}(\widetilde{W}_{j,k} \geq w)dw \\
&\int_{d_L}^{d_U} \int_{\mathcal{W}} d\mathbf{1}\{d \geq w\}dF_{\mathcal{W}}(d)dw \\
&\int_{\mathcal{W}} d\left(\int_{d_L}^{d_U} \mathbf{1}\{w \leq d\}dw\right) dF_{\mathcal{W}}(d) \\
&= \int_{\mathcal{W}} d(d-d_L)dF_{\mathcal{W}}(d) \\
&= \mathbb{E}[\widetilde{W}_{j,k}^2] - \mathbb{E}[\widetilde{W}_{j,k}]d_L
\end{aligned}
$$

where the first line is B_1, the second holds by the previous result, the third by changing the order of integration, the fourth by carrying out the inner integration, and the last by rewriting the integral as an expectation. Next, for B_2:

$$
\begin{aligned}
B_2 &= \mathbb{E}[\widetilde{W}_{j,k}] \int_{d_L}^{d_U} \mathbb{P}(\widetilde{W}_{j,k} \geq w)dw \\
&= \mathbb{E}[\widetilde{W}_{j,k}]\mathbb{P}(\widetilde{W}_{j,k} > 0) \int_{d_L}^{d_U} \mathbb{P}(\widetilde{W}_{j,k} \geq w|\widetilde{W}_{j,k} > 0)dw \\
&= \mathbb{E}[\widetilde{W}_{j,k}]\mathbb{P}(\widetilde{W}_{j,k} > 0) \int_{d_L}^{d_U} \int_{d_L}^{d_U} \mathbf{1}\{d \leq w\}dF_{\widetilde{W}_{j,k}|\widetilde{W}_{j,k}>0}(d)dw \\
&= \mathbb{E}[\widetilde{W}_{j,k}]\mathbb{P}(\widetilde{W}_{j,k} > 0) \int_{d_L}^{d_U} \left(\int_{d_L}^{d_U} \mathbf{1}\{d \leq w\}dw\right) dF_{\widetilde{W}_{j,k}|\widetilde{W}_{j,k}>0}(d) \\
&= \mathbb{E}[\widetilde{W}_{j,k}]\mathbb{P}(\widetilde{W}_{j,k} > 0) \int_{d_L}^{d_U} (d-d_L)dF_{\widetilde{W}_{j,k}|\widetilde{W}_{j,k}>0}(d) \\
&= \mathbb{E}[\widetilde{W}_{j,k}]\mathbb{P}(\widetilde{W}_{j,k} > 0)(\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} > 0] - d_L) \\
&= \mathbb{E}[\widetilde{W}_{j,k}]^2 - \mathbb{E}[\widetilde{W}_{j,k}]\mathbb{P}[\widetilde{W}_{j,k} > 0]d_L
\end{aligned}
$$

where the first equality is the definition of B_2, the second equality holds by the law of iterated expectations, the third equality holds by writing $\mathbb{P}(\widetilde{W}_{j,k} \geq w|\widetilde{W}_{j,k} > 0)$ as an integral, the fourth equality changes the order of integration, the fifth equality carries out the inside integration, the sixth equality rewrites the integral as an expectation, the last equality holds by combining terms and by the law of iterated expectations. Then for B_3:

$$
\begin{aligned}
B_3 =& \mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} > 0]\mathbb{P}[\widetilde{W}_{j,k} > 0]d_L \\
&= \mathbb{E}[\widetilde{W}_{j,k}]d_L
\end{aligned}
$$

which holds by the law of iterated expectations. Finally, for B_4:

$$B_4 = \mathbb{E}[\widetilde{W}_{j,k}]\mathbb{P}[\widetilde{W}_{j,k} > 0]d_L$$

Hence: $B_1 - B_2 + B_3 + B_4 = \mathbb{E}[\widetilde{W}_{j,k}^2] - \mathbb{E}[\widetilde{W}_{j,k}]^2 = var(\widetilde{W}_{j,k})$, which means that $\int_{d_L}^{d_U} w_1(w)dw + w_0 = 1$. For this reason,

$$\gamma_{j,k} = \int_{d_L}^{d_U} q_1(w_{j,k}) \left(\frac{\delta \mathbb{E}[\widetilde{Y}_j(w_{j,k})|\widetilde{W}_{j,k} = w_{j,k}]}{dw_{j,k}} + \frac{\delta \mathbb{E}[\widetilde{Y}_j(w_{j,k})|\widetilde{W}_{j,k} = w_{j,k}] - \mathbb{E}[\widetilde{Y}_j(a)|\widetilde{W}_{j,k} = 0]}{\delta a} \Big|_{a=w_{j,k}} \right) dw_{j,
$$
$$+ q_0 \frac{\mathbb{E}[\widetilde{Y}_j(d_L)|\widetilde{W}_{j,k} = d_L] - \mathbb{E}[\widetilde{Y}_j(0)|\widetilde{W}_{j,k} = 0]}{d_L}$$

where

$$q_1(w_{j,k}) := \frac{(\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} \geq w_{j,k}] - \mathbb{E}[\widetilde{W}_{j,k}])\mathbb{P}(\widetilde{W}_{j,k} \geq w_{j,k})}{\text{var}(\widetilde{W}_{j,k})},$$
$$q_0 := \frac{(\mathbb{E}[\widetilde{W}_{j,k}|\widetilde{W}_{j,k} > 0] - \mathbb{E}[\widetilde{W}_{j,k}])\mathbb{P}(\widetilde{W}_{j,k} > 0)d_L}{\text{var}(\widetilde{W}_{j,k})}.$$

and the weights satisfy $\int_{d_L}^{d_U} q_1(w_{j,k})dw_{j,k} + q_0 = 1$. □

Proof. **Proof of Theorem 5**: Notice that

$$\gamma_{j,k} = \frac{cov(\widetilde{X}_k, \widetilde{X}_{k+j})}{var(\widetilde{X}_k)} = \frac{\sum_{t=1}^{T}(\widetilde{W}_{j,k,t} - \bar{W}_k)(\widetilde{Y}_{j,t}^1 - \widetilde{Y}_{j,t}^0 - \bar{Y}_j)}{\sum_{t=1}^{T}(\widetilde{W}_{j,k,t} - \bar{W}_k)^2}$$

where $\bar{Y}_j = \frac{1}{T}\sum_{t=1}^{T} \widetilde{Y}_{j,t}^1 - \widetilde{Y}_{j,t}^0$ and $\bar{W}_k = \frac{1}{T}\sum_{t=1}^{T} \widetilde{W}_{j,k,t}$. Moreover, since $\widetilde{W}_{j,k}$ is a dummy variable, we can define $\bar{Y}_{j,k}^{\text{treatment time}} = \sum_{t:W_{k,t}=1} \widetilde{Y}_{j,t}^1 - \widetilde{Y}_{j,t}^0 / (\sum t : \widetilde{W}_{j,k,t} = 1)$ and $\bar{Y}_{j,k}^{\text{control time}} = \sum_{t:W_{k,t}=0} \widetilde{Y}_{j,t}^1 - \widetilde{Y}_{j,t}^0 / (\sum t : W_{k,t} = 0)$, where $t : \widetilde{W}_{j,k,t} = 1$ indexes the events in which the dummy variable errors indicate the event 1. Hence, there exists an equivalence between the parametric formulation resulting from the VAR and the semi-parametric one traditionally described in **?** through $\gamma_{j,k} = \bar{Y}_{j,k}^{\text{treatment time}} - \bar{Y}_{j,k}^{\text{control time}}$. This means that the parameter $\gamma_{j,k}$ estimates the difference between the innovations in treated and control times. Moreover, notice that $\mathbb{E}[\widetilde{Y}_{j,t}^1 - \widetilde{Y}_{j,t}^0] = \mathbb{E}[Y_{j,t}^1 - Y_{j,t}^0] - \mathbb{E}[Y_{j,t}^1 - Y_{j,t}^0 | Y_{j,t-1,j}^1 - Y_{j,t-1}^0, ...]$. Hence, $\bar{Y}_j^{\text{treatment time}}$ represents the mean of the difference $Y_{j,t}^1 - Y_{j,t}^0$ in treated times after conditioning on the past and $\bar{Y}_j^{\text{control time}}$ represents the same difference in control times. This means that $\gamma_{j,k}$ essentially captures six differences, as follows

$$\begin{aligned}\gamma_{j,k} =&\mathbb{E}[Y^1_{j,t}|\widetilde{W}_{j,k,t}=1]-\mathbb{E}[Y^0_{j,t}|\widetilde{W}_{j,k,t}=1]-\mathbb{E}[Y^1_{j,t}-Y^0_{j,t}|\widetilde{W}_{j,k,t}=1,,Y^1_{j,t-1,j}-Y^0_{j,t-1},...]\\ &-\mathbb{E}[Y^1_{j,t}|\widetilde{W}_{j,k,t}=0]+\mathbb{E}[Y^0_{j,t}|\widetilde{W}_{j,k,t}=0]+\mathbb{E}[Y^1_{j,t}-Y^0_{j,t}|\widetilde{W}_{j,k,t}=0,Y^1_{j,t-1,j}-Y^0_{j,t-1},...]\end{aligned} \tag{11}$$

Define $\Delta_{ATT} = \mathbb{E}[Y^1_{j,t}(1)|\widetilde{W}_{j,k,t} = 1] - \mathbb{E}[Y^0_{j,t}(0)|\widetilde{W}_{j,k,t} = 1] - \mathbb{E}[Y^1_{j,t}(0)|\widetilde{W}_{j,k,t} = 0] + \mathbb{E}[Y^0_{j,t}(0)|\widetilde{W}_{j,k,t} = 0]$ and $\Delta_{AR} = \mathbb{E}[Y^1_{j,t} - Y^0_{j,t}|\widetilde{W}_{j,k,t} = 0, Y^1_{j,t-1} - Y^0_{j,t-1}, ...] - \mathbb{E}[Y^1_{j,t} - Y^0_{j,t}|\widetilde{W}_{j,k,t} = 1, , Y^1_{j,t-1} - Y^0_{j,t-1}, ...]$ Using the definition of potential outcomes in 11, equation 11 becomes.

$$\begin{aligned}\Delta_{ATT} =&\mathbb{E}[Y^1_{j,t}(1)|\widetilde{W}_{j,k,t}=1]-\mathbb{E}[Y^0_{j,t}(0)|\widetilde{W}_{j,k,t}=1]\\ &-\mathbb{E}[Y^1_{j,t}(0)|\widetilde{W}_{j,k,t}=0]+\mathbb{E}[Y^0_{j,t}(0)|\widetilde{W}_{j,k,t}=0]\end{aligned}$$

Adding and subtracting $\mathbb{E}[Y^1_{j,t}(0)|\widetilde{W}_{j,k,t} = 1]$ it becomes

$$\begin{aligned}\Delta_{ATT} =&\mathbb{E}[Y^1_{j,t}(1)|\widetilde{W}_{j,k,t}=1]-\mathbb{E}[Y^1_{j,t}(0)|\widetilde{W}_{j,k,t}=1]\\ &-\mathbb{E}[Y_{j,t}(0)|\widetilde{W}_{j,k,t}=0]+\mathbb{E}[Y^0_{j,t}(0)|\widetilde{W}_{j,k,t}=0]\\ &+\mathbb{E}[Y_{j,t}(0)|\widetilde{W}_{j,k,t}=1]-\mathbb{E}[Y^0_{j,t}(0)|\widetilde{W}_{j,k,t}=1]\end{aligned}$$

by using assumptions 13 and 14 it becomes

$$\Delta_{ATT} =\mathbb{E}[Y^1_{j,t}(1)|\widetilde{W}_{j,k,t}=1]-\mathbb{E}[Y^1_{j,t}(0)|\widetilde{W}_{j,k,t}=1]$$

Then, using the definition of the outcome process from assumption 11 it becomes $\Delta_{ATT} = \mathbb{E}[Y_{j,t}(1) - Y_{j,t}(0)|\widetilde{W}_{j,k,t} = 1]$. Finally, Δ_{AR} remains as a residual that is zero when the difference between the predicted autoregressive component of the treated and control is zero. □

Proof. **Proof of Theorem 6.** Start from the result of Theorem 2. Then,

$$\gamma_{jk} = \frac{cov(\widetilde{Y}_j, \widetilde{W}_{j,k})}{var(\widetilde{W}_{j,k})} = \frac{\delta \mathbb{E}[\widetilde{Y}_j | \widetilde{W}_{j,k,t} = w_{j,k}]}{\delta w_{j,k}}$$

Now considering

$$\begin{aligned}
\gamma_{j,k} &= \frac{\delta \mathbb{E}[\widetilde{Y}_j | \widetilde{W}_{j,k,t} = w_{j,k}]}{\delta w_{j,k}} \\
&= \frac{\delta}{\delta w_{j,k}} \Big(\mathbb{E}[Y_t^1 - Y_t^0 | \widetilde{W}_{j,k,t} = w_{j,k}] \\
&\quad - \mathbb{E}[Y_t^1 | \widetilde{W}_{j,k,t} = w_{j,k}, Y_{t-1}^0, Y_{t-1}^1, \dots] \\
&\quad + \mathbb{E}[Y_t^0 | \widetilde{W}_{j,k,t} = w_{j,k}, Y_{t-1}^0, Y_{t-1}^1, \dots] \Big) \\
&= \frac{\delta \mathbb{E}[Y_t^1 - Y_t^0 | \widetilde{W}_{j,k} = 1]}{\delta w_{j,k}} + \Delta_{AR} \\
&= \frac{\delta \mathbb{E}[Y_t^1(w_{j,k}) - Y_t^0(0) | \widetilde{W}_{j,k} = 1]}{\delta w_{j,k}} + \Delta_{AR} \\
&= \frac{\delta \mathbb{E}[Y_t(w_{j,k}) - Y(0)]}{\delta w_{j,k}} + \Delta_{AR}
\end{aligned}$$

where the first equality holds by the definition of the VAR, the second holds by the linearity property of expectations, the third holds by substituting the potential outcome definition in 15, and the fourth holds by the strong parallel trend assumption 16. □

Proof. **Proof of Theorem 7.** The proof is akin to the one of **?**. □

Proof. **Proof of Theorem 8.** By the same type of argument of Theorem 5, $\gamma_{j,k}$ becomes

$$\begin{aligned}
\gamma_{j,k} =& \mathbb{E}[\Delta Y_{j,t} | \widetilde{W}_{j,k,t} = 1] - \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^p A_l \Delta X_{t-l}) | \widetilde{W}_{j,k,t} = 1] \\
& - \mathbb{E}[\Delta Y_{j,t} | \widetilde{W}_{j,k,t} = 0] + \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^p A_l \Delta X_{t-l}) | \widetilde{W}_{j,k,t} = 0]
\end{aligned}$$

Moving to the potential outcome notation, this means that

$$\begin{aligned}
\gamma_{j,k} =& \mathbb{E}[\Delta Y_{j,t}(1) | \widetilde{W}_{j,k,t} = 1] - \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^p A_l \Delta X_{t-l}) | \widetilde{W}_{j,k,t} = 1] \\
& - \mathbb{E}[\Delta Y_{j,t}(0) | \widetilde{W}_{j,k,t} = 0] + \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^p A_l \Delta X_{t-l}) | \widetilde{W}_{j,k,t} = 0]
\end{aligned}$$

Adding and subtracting $\mathbb{E}[\Delta Y_{j,t}(0) | \widetilde{W}_{j,k,t} = 1]$ it becomes

$$
\begin{aligned}
\gamma_{j,k} =& \mathbb{E}[\Delta Y_{j,t}(1)|\widetilde{W}_{j,k,t}=1] - \mathbb{E}[\Delta Y_{j,t}(0)|\widetilde{W}_{j,k,t}=1] \\
& - \mathbb{E}[\Delta Y_{j,t}(0)|\widetilde{W}_{j,k,t}=0] + \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^{p} A_l \Delta X_{t-l})|\widetilde{W}_{j,k,t}=0] \\
& + \mathbb{E}[\Delta Y_{j,t}(0)|\widetilde{W}_{j,k,t}=1] - \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^{p} A_l \Delta X_{t-l})|\widetilde{W}_{j,k,t}=1]
\end{aligned}
$$

By assumptions 18 and 19 the second and third line cancel out, returning the result in the theorem. □

Proof. **Proof of Theorem 9.** Start from the result of Theorem 2. Then,

$$
\gamma_{jk} = \frac{cov(\widetilde{Y}_j, \widetilde{W}_{j,k})}{var(\widetilde{W}_{j,k})} = \frac{\delta \mathbb{E}[\widetilde{Y}_j|\widetilde{W}_{j,k}=w_{j,k}]}{\delta w_{j,k}}.
$$

By using assumption 20, the numerator becomes

$$
\mathbb{E}[\widetilde{Y}_j|\widetilde{W}_{j,k,t}=w_{j,k}] = \mathbb{E}[\Delta Y^1_{j,t} - f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^{p} A_l \Delta X_{t-l})|\widetilde{W}_{j,k,t}=w_{j,k}]
$$

which becomes

$$
\mathbb{E}[\Delta Y^1_{j,t}(w_{j,k}) - f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^{p} A_l \Delta X_{t-l})|\widetilde{W}_{j,k,t}=w_{j,k}]
$$

by the definition of potential outcomes. Adding and subtracting $\mathbb{E}[\Delta Y_{j,t}(0)|\widetilde{W}_{j,k,t}=w_{j,k}]$ it yields

$$
\begin{aligned}
& \mathbb{E}[\Delta Y_{j,t}(w_{j,k})|\widetilde{W}_{j,k,t}=w_{j,k}] - \mathbb{E}[\Delta Y_{j,t}(0)|\widetilde{W}_{j,k,t}=w_{j,k}] \\
& \mathbb{E}[f_{K+j}(\Pi X_{t-1}) + f_{K+j}(\Sigma_{l=1}^{p} A_l \Delta X_{t-l})|\widetilde{W}_{j,k,t}=w_{j,k}] - \mathbb{E}[\Delta Y_{j,t}(0)|\widetilde{W}_{j,k,t}=w_{j,k}]
\end{aligned}
$$

where the second line cancels out by assumption 20. Hence, it remains

$$
\mathbb{E}[\Delta Y_{j,t}(w_{j,k}) - \Delta Y_{j,t}(0)|\widetilde{W}_{j,k,t}=w_{j,k}]
$$

which conditioning argument cancels out by assumption 21. □